NOBELPRISET I KEMI 2024
THE NOBEL PRIZE IN CHEMISTRY 2024

KUNGL. VETENSKAPS-AKADEMIEN
THE ROYAL SWEDISH ACADEMY OF SCIENCES


Photo: University of Washington

David Baker
University of Washington
USA

”för datorbaserad proteindesign”

“for computational protein design”


Photo: The Royal Society

Demis Hassabis
Google DeepMind
United Kingdom


Photo: BBVA Foundation

John M. Jumper
Google DeepMind
United Kingdom

”för proteinstrukturprediktion”

“for protein structure prediction”

#NobelPrize

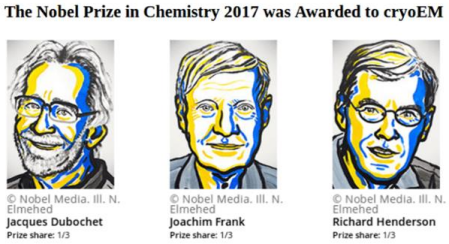
THE NOBEL PRIZE

LIVE 8,975 views



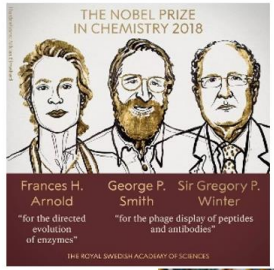


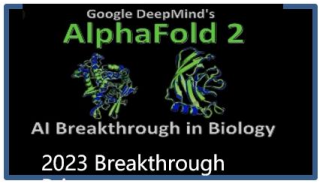




理解

Experimental structural biology





蛋白质结构数据库

人工智能的
源数据库





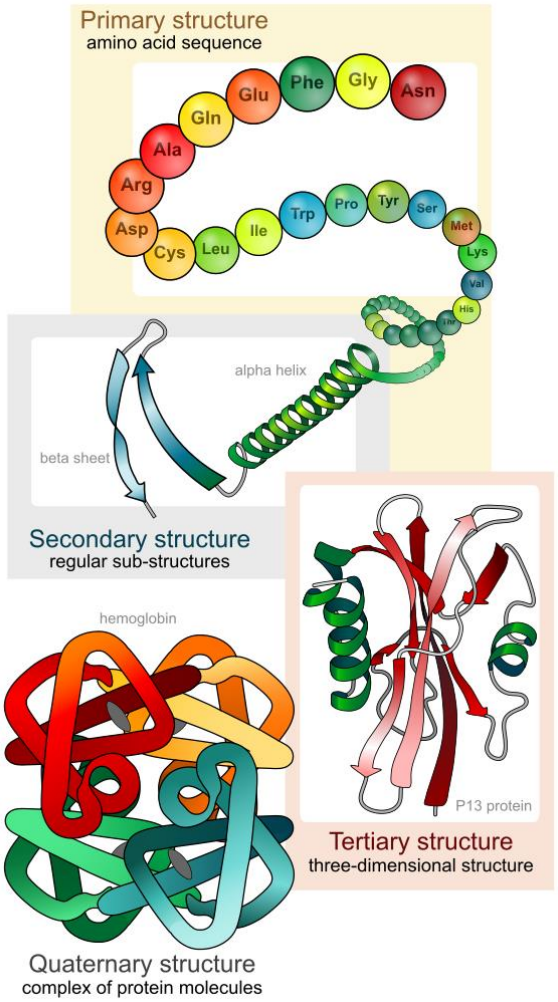
Computational structural modeling

模拟

Protein design
Drug design

创造,应用

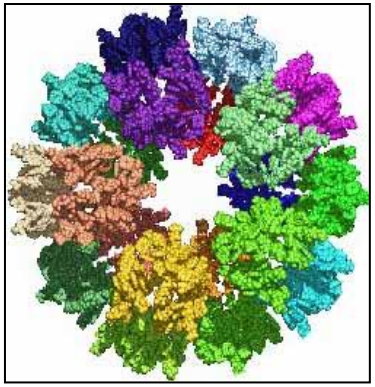
蛋白质3D结构决定其功能和调控机制



空间结构

蛋白修饰

相互作用



药物研发
疾病诊断
抗体 疫苗
生物技术
现代工农业

从基因（DNA）转录翻译而来的线性的蛋白质链折叠成特定的空间结构，然后具有功能；

蛋白质的空间结构决定其功能及作用机制，是药物设计的基础。

蛋白质结构

蛋白质的分子组成和化学结构

蛋白质的空间结构

蛋白质结构与功能的关系

蛋白质的理化性质

第一节

蛋白质的分子组成和化学结构

Molecular Component of Proteins

■ 蛋白质的生物学重要性

1. 蛋白质是生物体主要的功能执行者

2. 蛋白质具有多样性的生物学功能

1）作为生物催化剂——酶类 3000种以上

2）代谢调节作用—— 激素蛋白

3）免疫保护作用—— 抗体

4）物质的转运和存储——血红蛋白、脂蛋白

5）运动与支持作用——肌蛋白

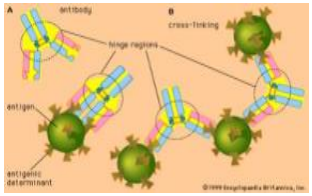
6）控制生长和分化作用——核蛋白、组蛋白

7）参与细胞间信息传递——跨膜蛋白

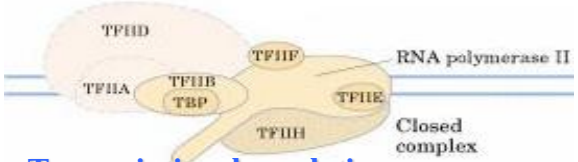
8）参与组成生物膜

3. 氧化供能

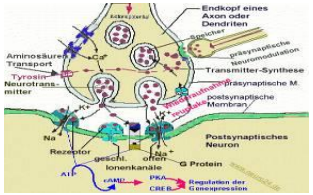
蛋白质的功能



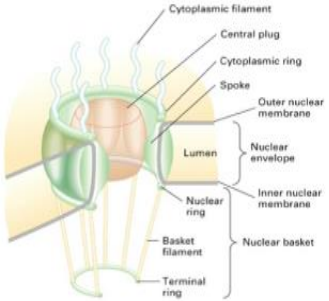
Immune response



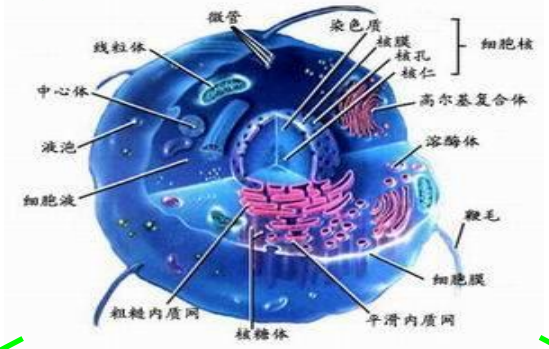
Transcriptional regulation

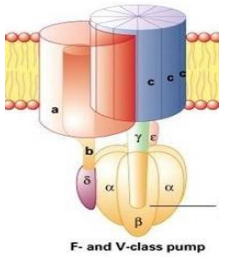


Signal transduction

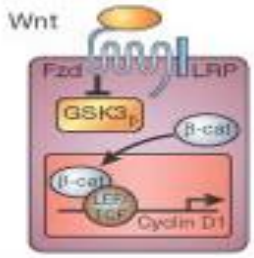


Material transportation

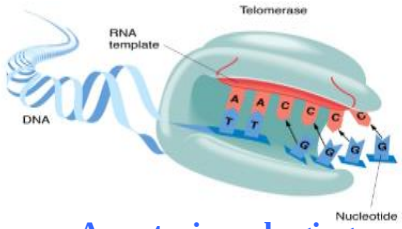




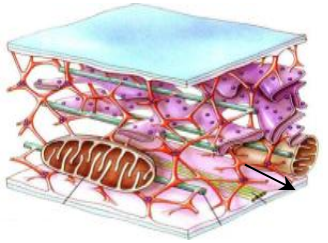
Metabolism



Intracellular signal pathway



Apoptosis and aging



Microfilament

■ 蛋白质的元素组成

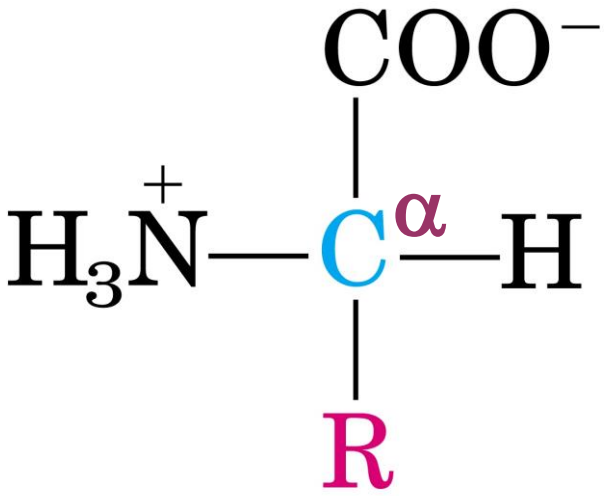
主要有C、H、O、N和 S。

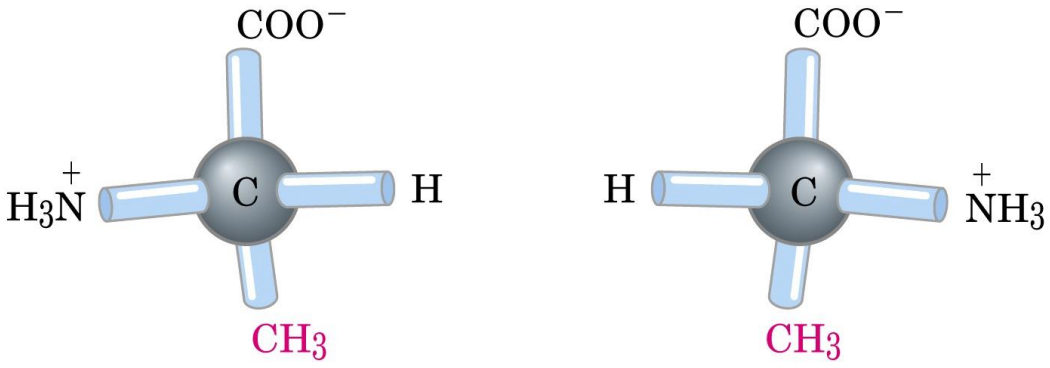
有些蛋白质含有少量磷或金属元素铁、铜、锌、锰、钴、钼，个别蛋白质还含有碘 。

蛋白质(protein)是由许多氨基酸通过肽键相连形成的高分子含氮化合物。 各种蛋白质的含N量很接近，平均为16%

一、蛋白质的基本组成单位——氨基酸

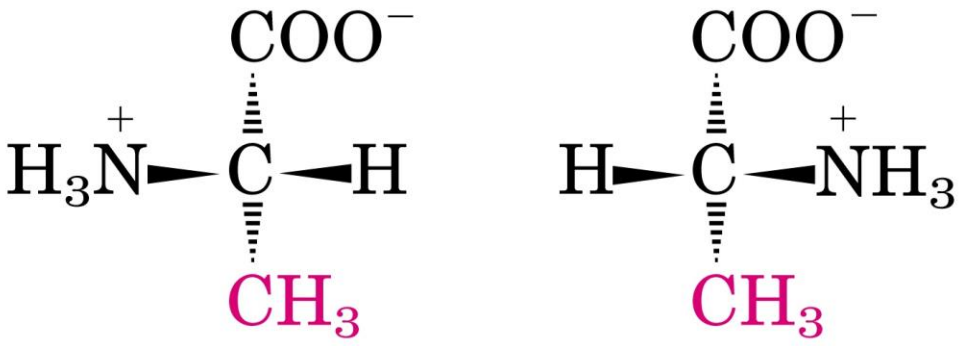
存在自然界中的氨基酸有300余种，但组成人体蛋白质的氨基酸仅有20种，且均属 L-α-氨基酸（甘氨酸除外）。





L-Alanine D-Alanine

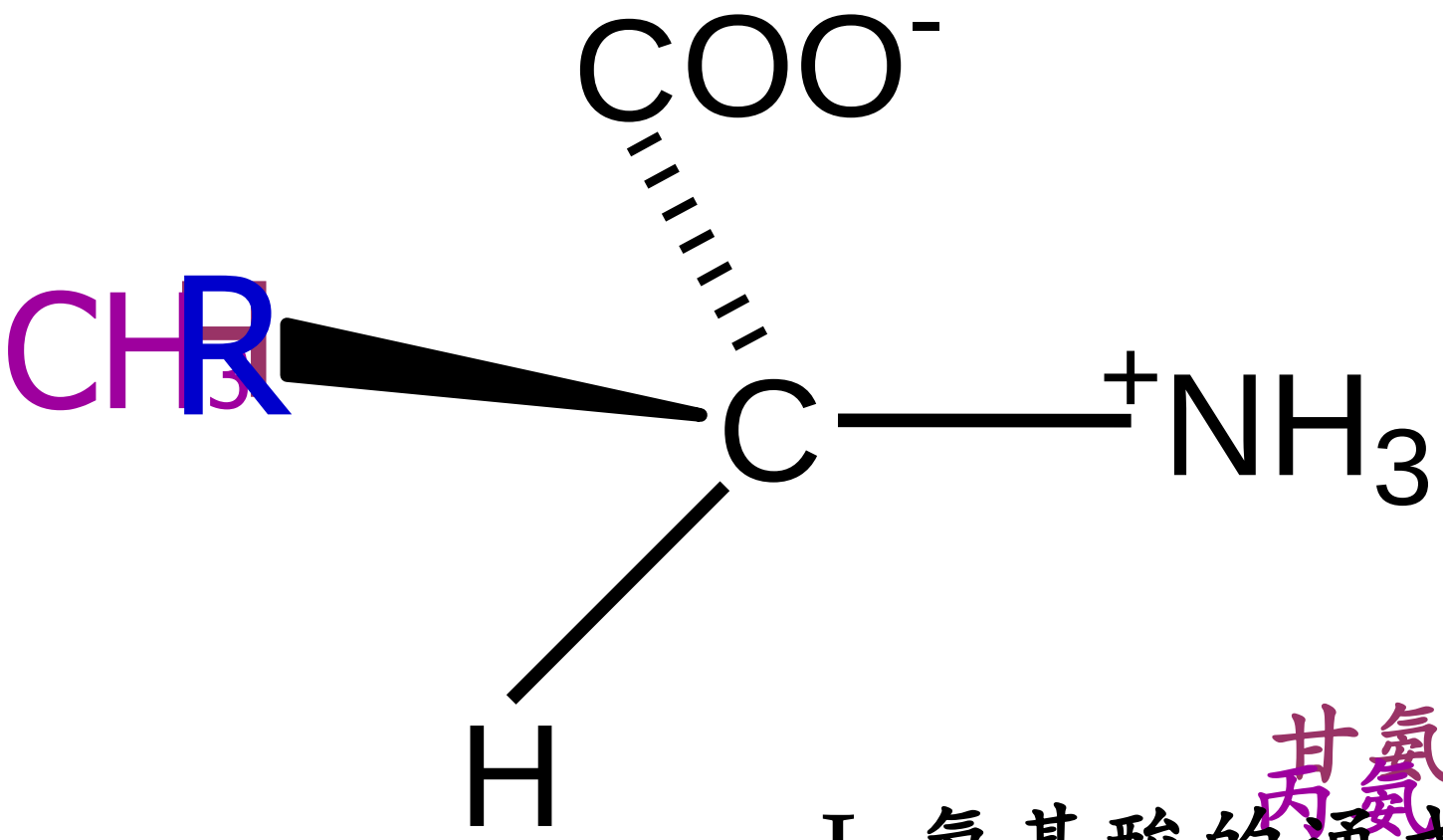
(a)



L-Alanine D-Alanine

(b)

二十种氨基酸的不同之处——R侧链



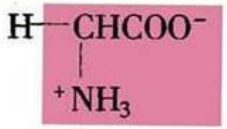
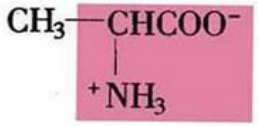
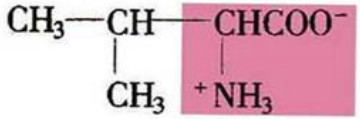
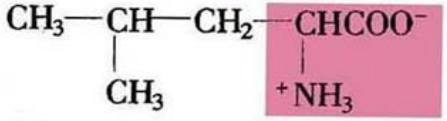
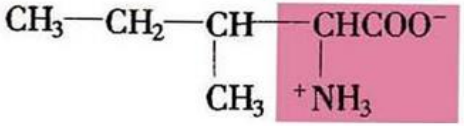
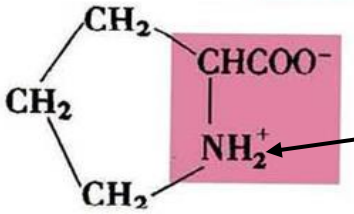
甘氨酸
丙氨酸

L-氨基酸的通式

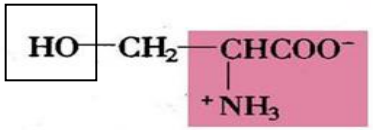
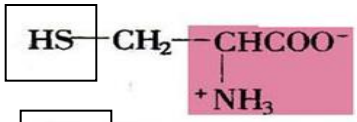
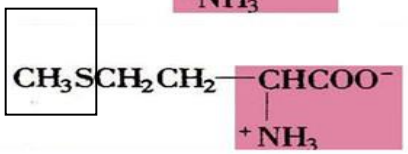
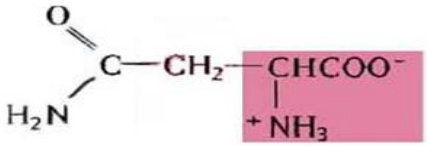
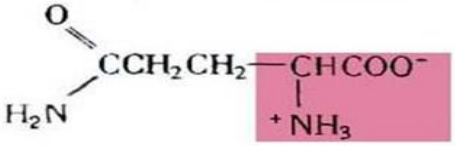
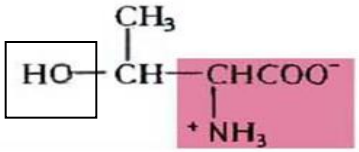
二、氨基酸可根据R侧链结构和理化性质进行分类

- 非极性脂肪族氨基酸
- 极性中性氨基酸
- 芳香族氨基酸
- 酸性氨基酸
- 碱性氨基酸

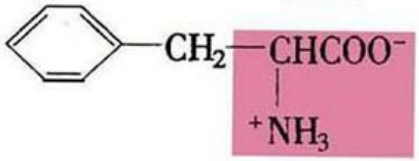
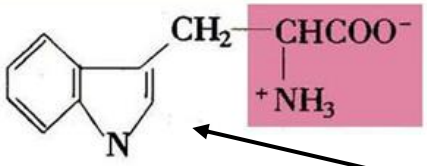
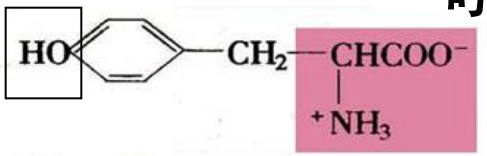
(一)侧链含烃链的氨基酸属于非极性脂肪族氨基酸

结构式	中文名	英文名	三字符号	一字符号	等电点(pI)
	甘氨酸	glycine	Gly	G	5.97
	丙氨酸	alanine	Ala	A	6.00
	缬氨酸	valine	Val	V	5.96
	亮氨酸	leucine	Leu	L	5.98
	异亮氨酸	isoleucine	Ile	I	6.02
 亚氨基	脯氨酸	proline	Pro	P	6.30

(二)侧链有极性但不带电荷的氨基酸是极性中性氨基酸

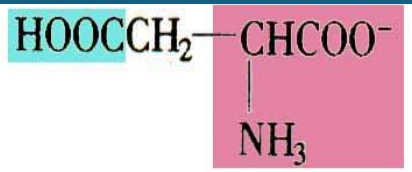
结构式	中文名	英文名	三字符号	一字符号	等电点(pI)
	丝氨酸	serine	Ser	S	5.68
	半胱氨酸	cysteine	Cys	C	5.07
	蛋氨酸（甲硫氨酸）	methionine	Met	M	5.74
	天冬酰胺	asparagine	Asn	N	5.41
	谷氨酰胺	glutamine	Gln	Q	5.65
	苏氨酸	threonine	Thr	T	5.60

(三)侧链含芳香基团的氨基酸是芳香族氨基酸

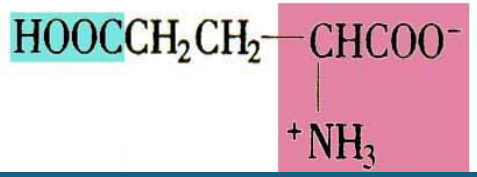
结构式	中文名	英文名	三字符号	一字符号	等电点(pI)
	苯丙氨酸	phenylalanine	Phe	F	5.48
	色氨酸	tryptophan	Try	W	5.89
	酪氨酸	tyrosine	Trp	Y	5.66

吲哚基

（四） 酸性氨基酸 （两个羧基）

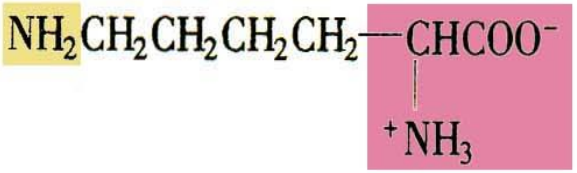


天冬氨酸 aspartic acid **Asp** D 2.97

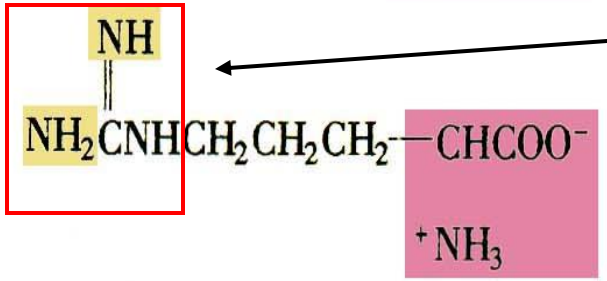


谷氨酸 glutamic acid **Glu** E 3.22

（五） 碱性氨基酸 （两个以上氨基）

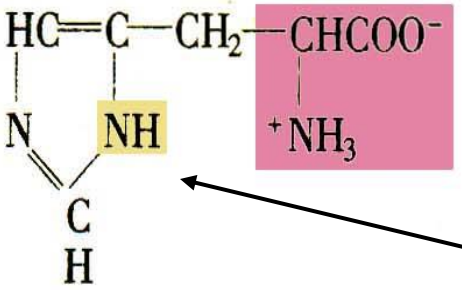


赖氨酸 lysine Lys K 9.74



胍基

精氨酸 arginine Arg R 10.76



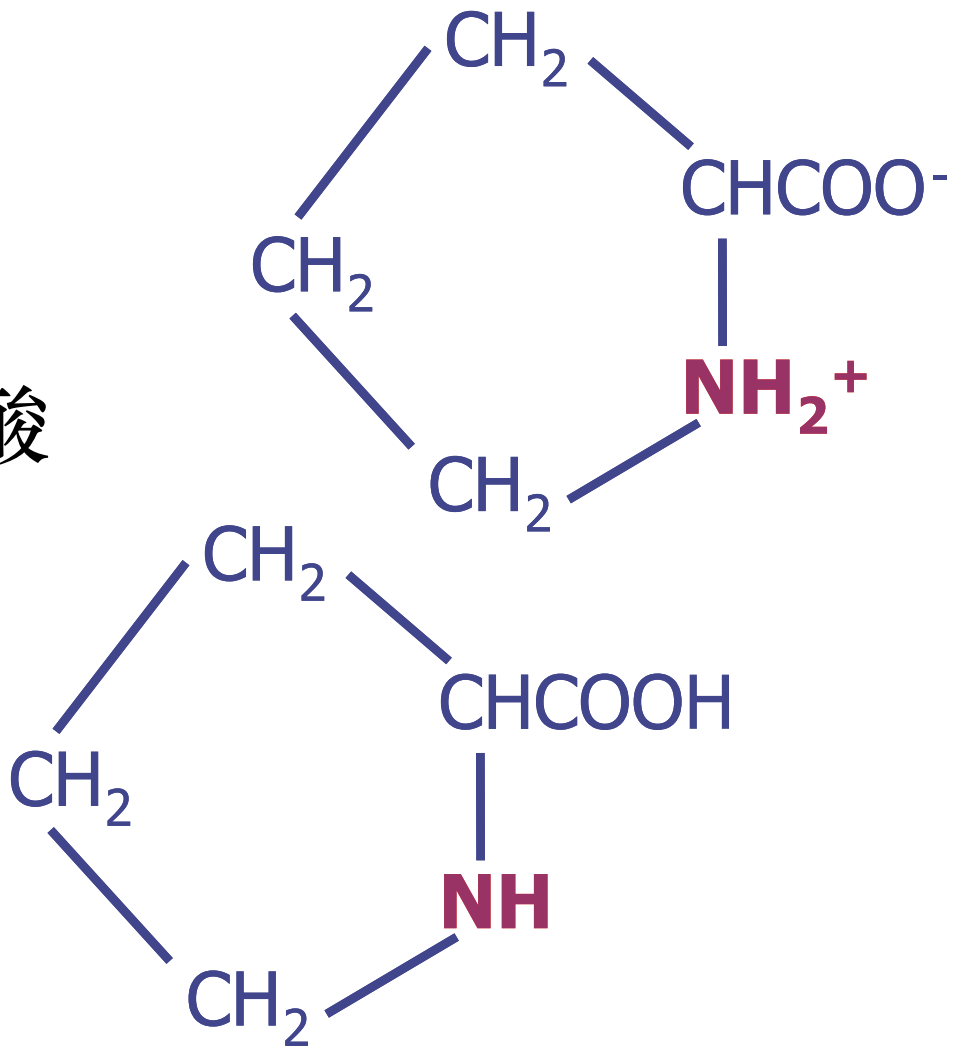
组氨酸 histidine His H 7.59

咪唑基

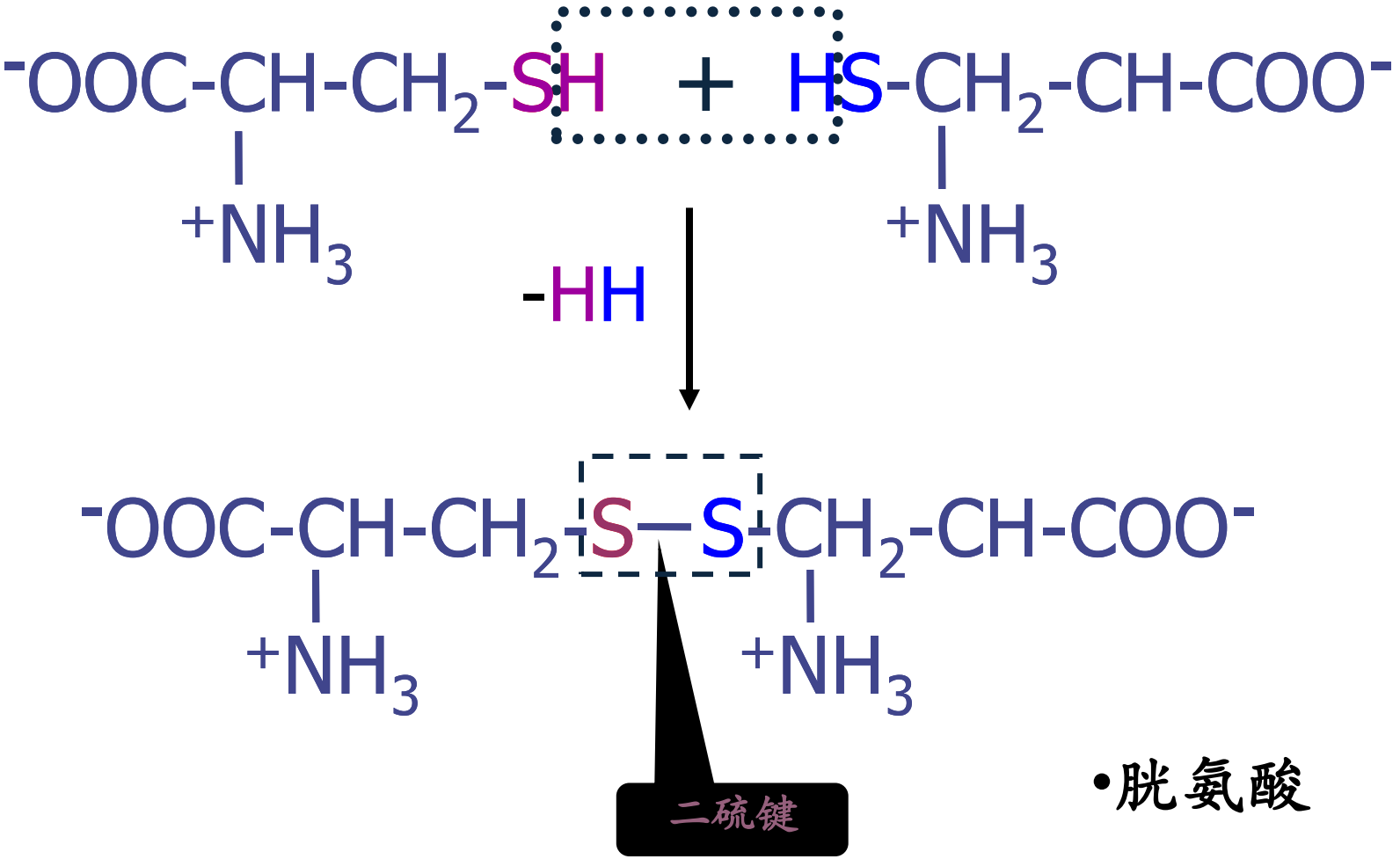
■ 几种特殊氨基酸

甘：无手性碳原子

脯：为环状亚氨基酸



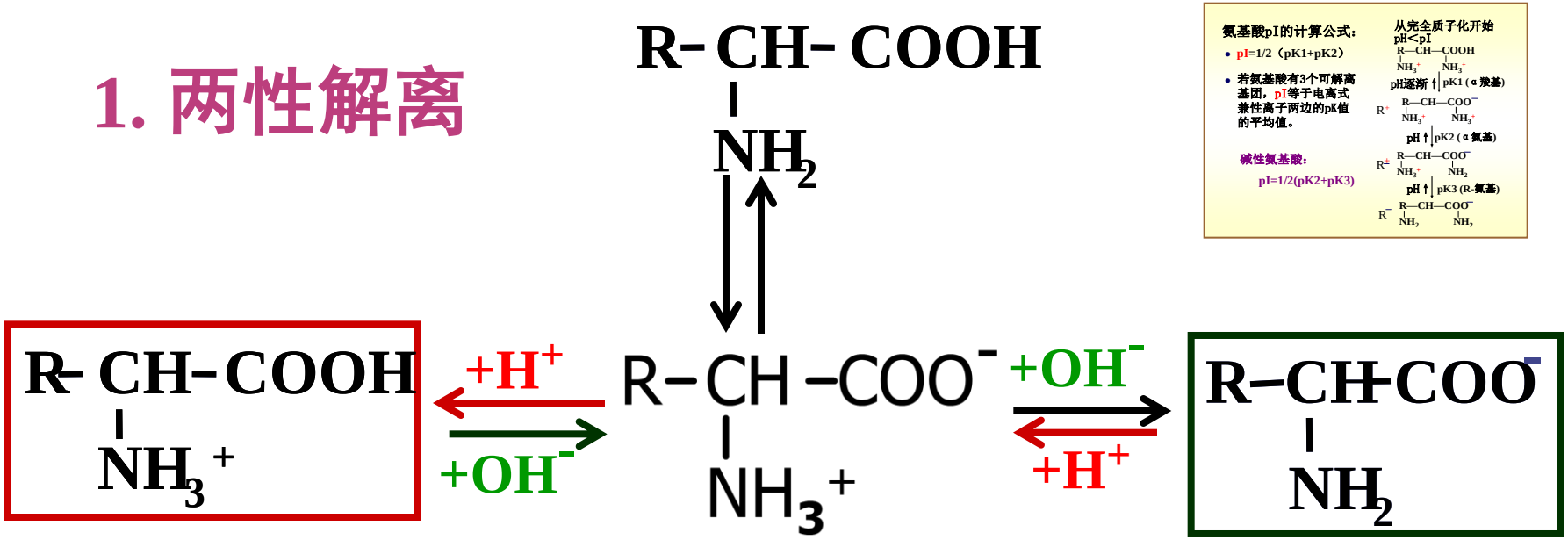
- 半胱氨酸


-OOC-CH-CH2-SH + HS-CH2-CH-COO-
+NH3
+NH3
-HH
-OOC-CH-CH2-S—S-CH2-CH-COO-
+NH3
+NH3
二硫键

•胱氨酸

三、氨基酸的理化性质

1. 两性解离



两性电解质，其解离程度取决于所处溶液的酸碱度。

等电点(isoelectric point, pI)

等于特定条件下的pH值

氨基酸在某一pH的溶液中，解离成阳离子和阴离子的趋势及程度相等，成为兼性离子，呈电中性，此时溶液的pH值称为该氨基酸的等电点。

氨基酸pI的计算公式：

- pI=1/2（pK1+pK2）
- 若氨基酸有3个可解离基团，pI等于电离式兼性离子两边的pK值的平均值。

碱性氨基酸：

pI=1/2(pK2+pK3)

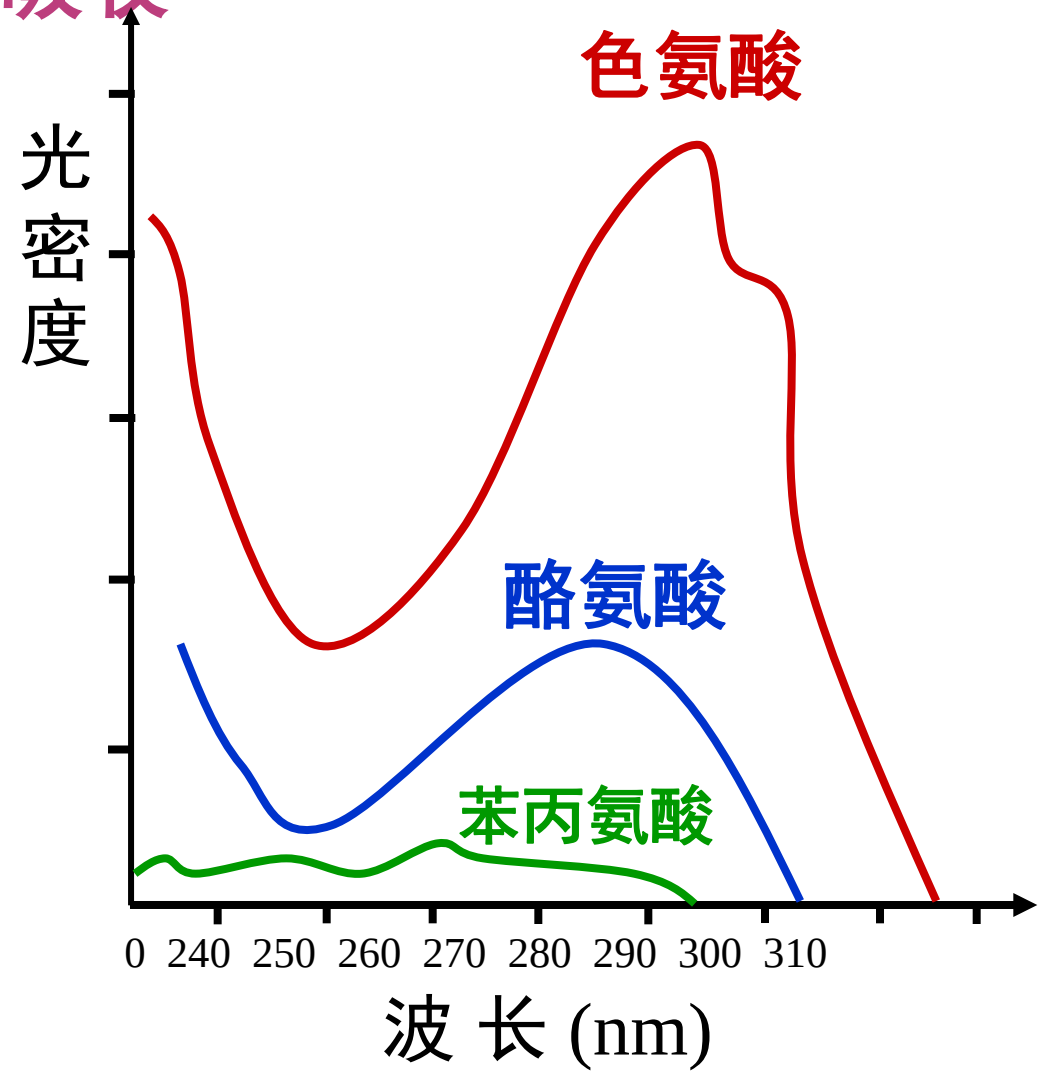
从完全质子化开始 pH<pI

2.芳香族氨基酸紫外吸收

色氨酸、酪氨酸

多数蛋白含这两种氨基酸残基，测定蛋白溶液280nm光吸收值可快速简便分析蛋白含量。

另外，肽键在短波长紫外区有强烈吸收。

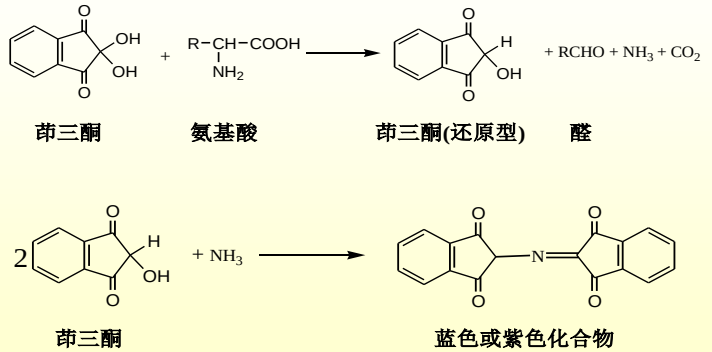


芳香族氨基酸共轭基团的吸收光谱

3. 茚三酮反应

•原理：氨基酸与茚三酮水合物共热，可生成蓝紫色化合物，其最大吸收峰在570nm处。

茚三酮反应过程



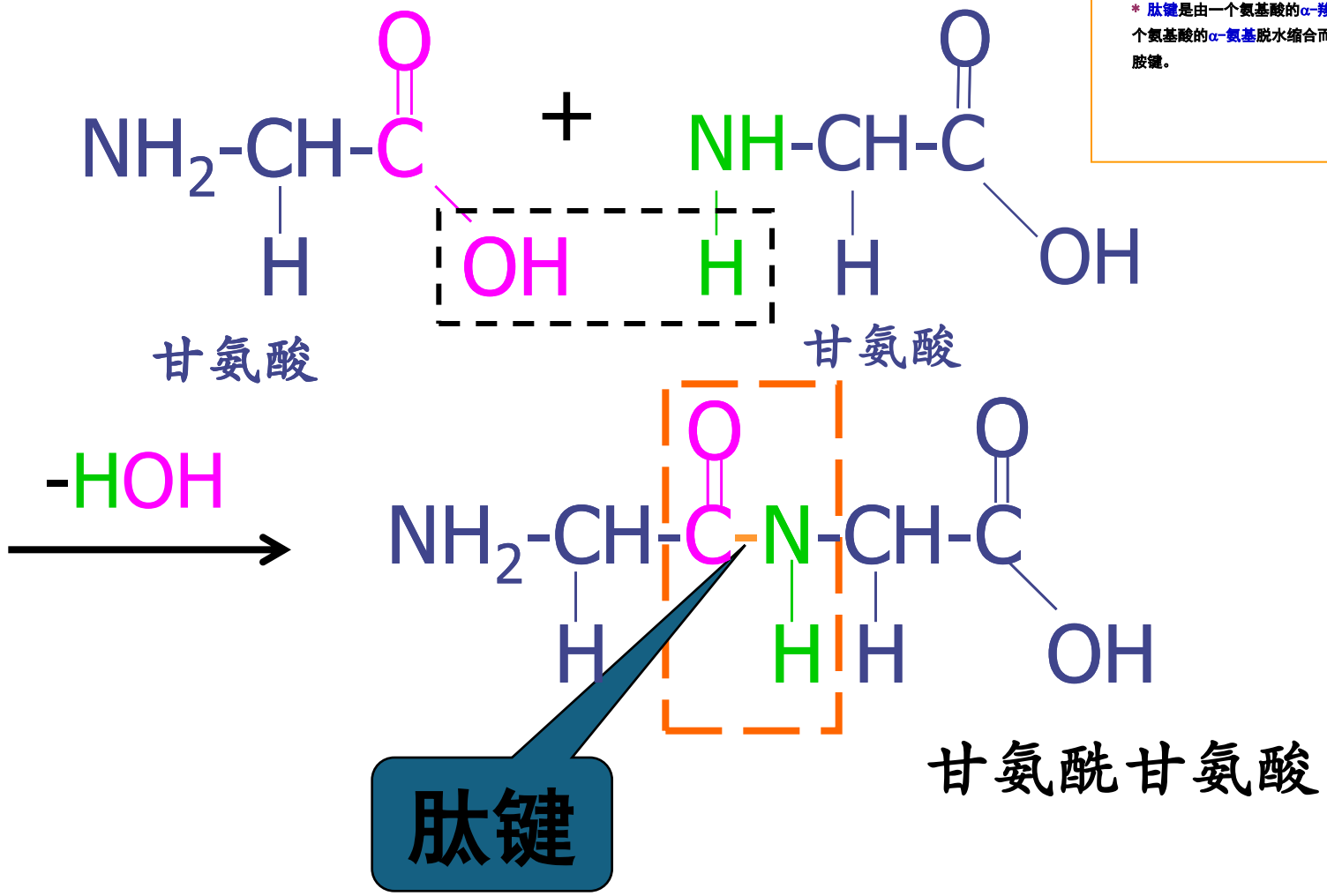
• 应用：此法可以对氨基酸进行定量分析（吸收峰值与氨基酸的含量存在正比关系）。

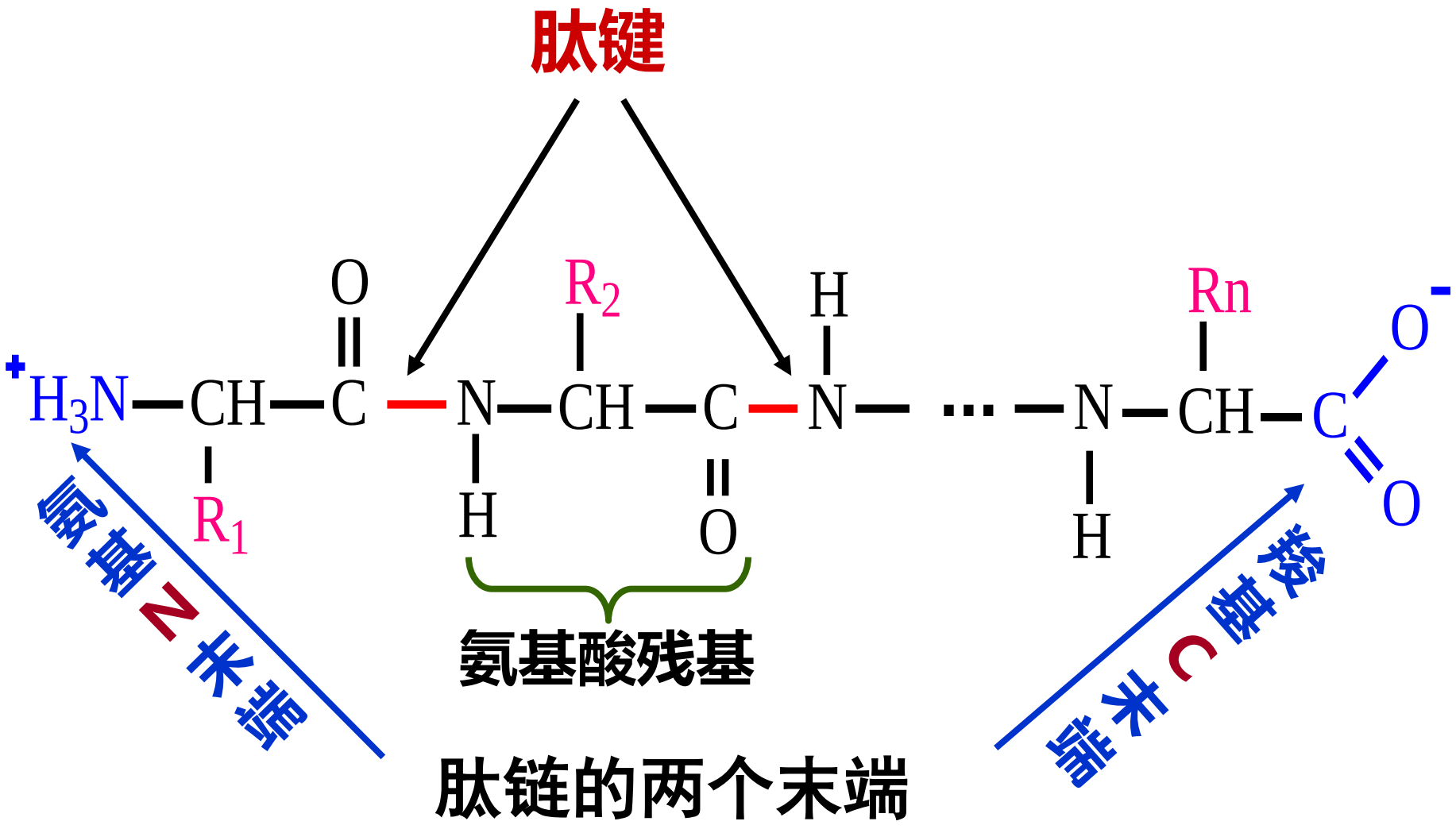
四、蛋白质是由许多氨基酸残基组成的多肽链

（一）肽（peptide)

* 肽键是由一个氨基酸的α-羧基与另一个氨基酸的α-氨基脱水缩合而形成的酰胺键。

肽键的形成




肽键
氨基酸残基
氨基N末端
羧基C末端
肽链的两个末端

肽是由氨基酸通过肽键缩合而形成的化合物。

两分子氨基酸缩合形成二肽，三分子氨基酸缩合则形成三肽……

由十个以内氨基酸相连而成的肽称为寡肽

由更多的氨基酸相连形成的肽称多肽

肽链中的氨基酸分子因为脱水缩合而基团不全，被称为氨基酸残基

R基团在链的两侧，故称为R侧链

- 多肽链有两端：

N 末端：多肽链中有游离α-氨基的一端

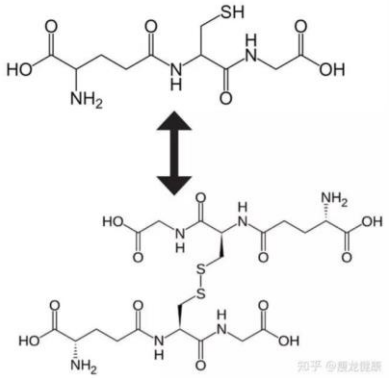
C 末端：多肽链中有游离α-羧基的一端

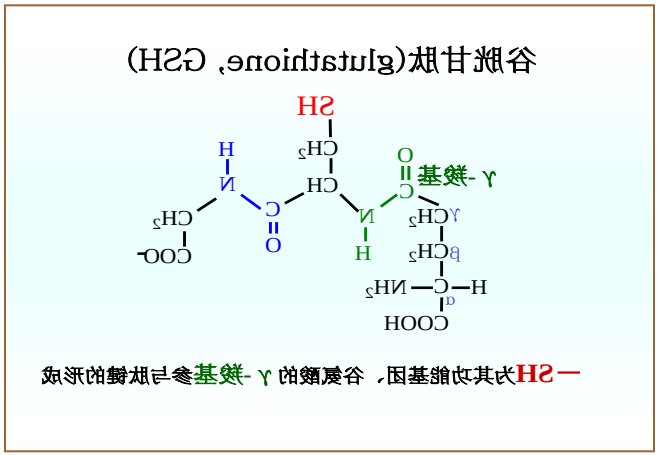
（二）生物活性肽

人体内具有特殊生物活性的肽类，如：

- **谷胱甘肽（glutathione，GSH）**

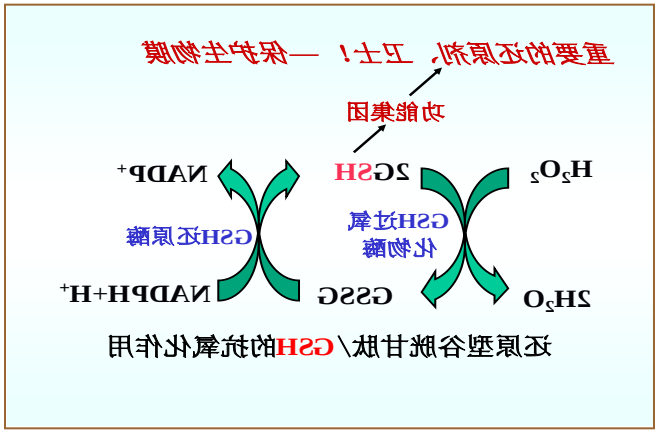
➢谷胱甘肽的结构





➢谷胱甘肽参加的反应及功能

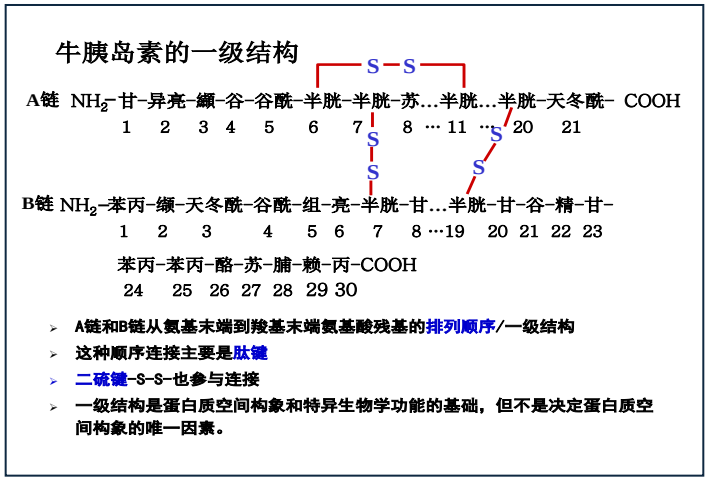
谷胱甘肽，是人体**本身存在的抗氧化剂**，是货真价实的自由基“收割机”，能减少氧化应激带来的损伤。GSH与GSSG比值决定细胞氧化还原状态，静止状态下的健康细胞GSH/GSSG比值＞100，而在氧化应激下细胞比例下降到1～10。

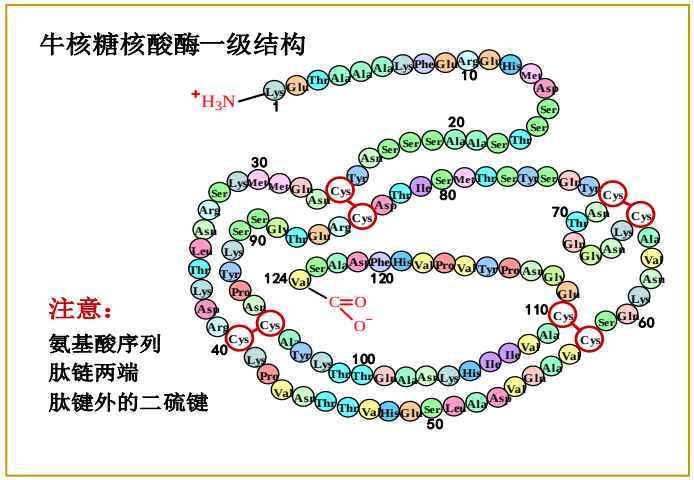


蛋白质的一级结构(primary structure)

- 什么是?

多肽链中从N→C端氨基酸的排列顺序。





- 主要的化学键— 肽键

有些蛋白质还包括二硫键

蛋白质序列（一级结构）数据库与蛋白质修饰

- 目前已知一级结构的蛋白质数量已相当可观，并且还以更快的速度增加。

  国际互联网有若干重要的蛋白质数据库，例如

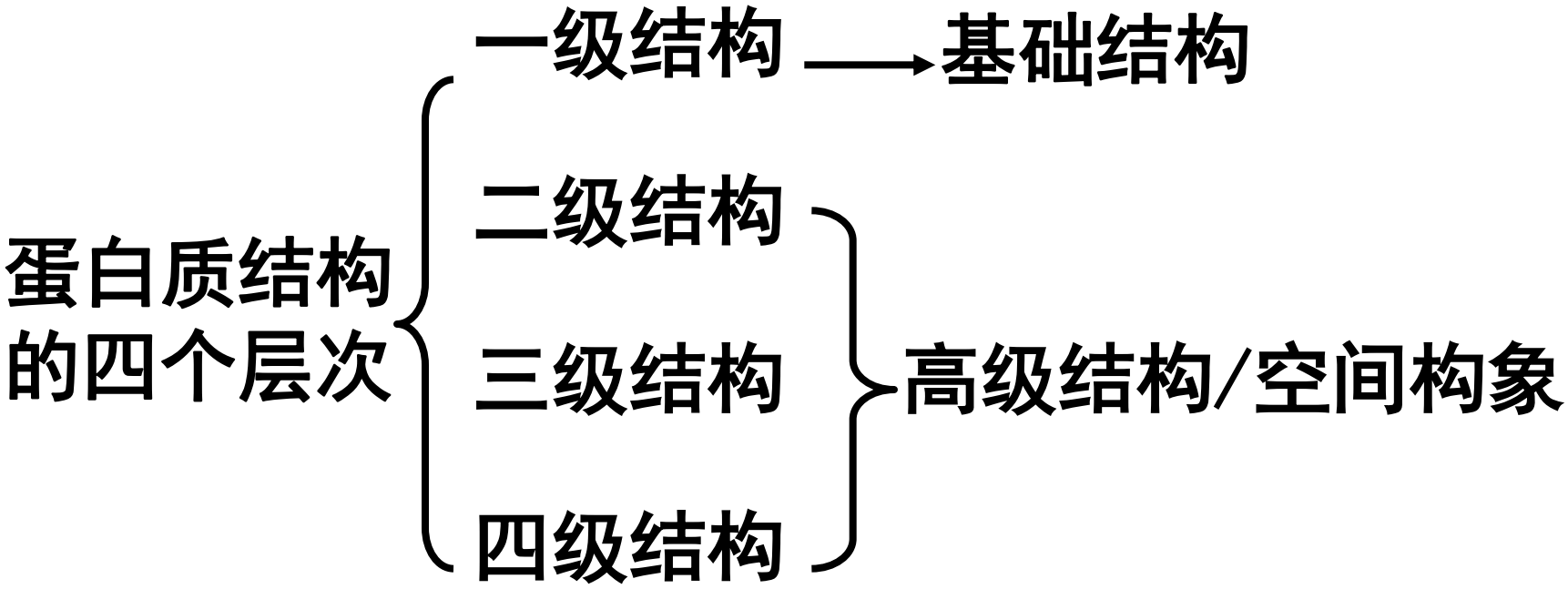
  - EMBL（European Molecular Biology Laboratory Data Library）
  - Uniprot (http://www.uniprot.org)
  - Genbank （Genetic Sequence Databank）
  - PIR（Protein Identification Resource Sequence Database）

  收集了大量最新的蛋白质一级结构及其他资料，为蛋白质结构与功能的深入研究提供了便利。

第二节

蛋白质的空间结构

The Molecular Structure of Protein


蛋白质结构
的四个层次
一级结构
基础结构
二级结构
三级结构
四级结构
高级结构/空间构象

一、蛋白质的一级结构(primary structure)

- **一级结构的阐明具有重要意义：**

一级结构的重要意义：

1、20种氨基酸排列顺序不同而组成多种多样的蛋白质，并具有不同的生物学功能。

2、许多先天性疾病是由于某一重要蛋白一级结构改变而引起。

3、蛋白质一级结构中氨基酸排列顺序是由遗传密码所决定，氨基酸的改变是碱基顺序改变所致。

- 一级结构是蛋白质空间构象和特异生物学功能的基础，但不是决定蛋白质空间构象的唯一因素。

二、蛋白质的二级结构(secondary structure)

- **什么是？ (关键词:主链、肽段、局部)**

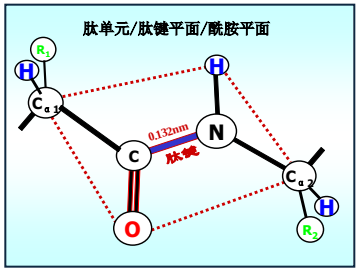
  蛋白质分子中某一段肽链的局部空间结构，即该段肽链主链骨架原子的相对空间位置，不涉及R侧链的构象 。

- **二级结构的主要化学键—氢键**

- **二级结构的基本单位— 肽单元**

（一）二级结构的基础—肽单元(peptide unit)

- **对肽单元的说明：**



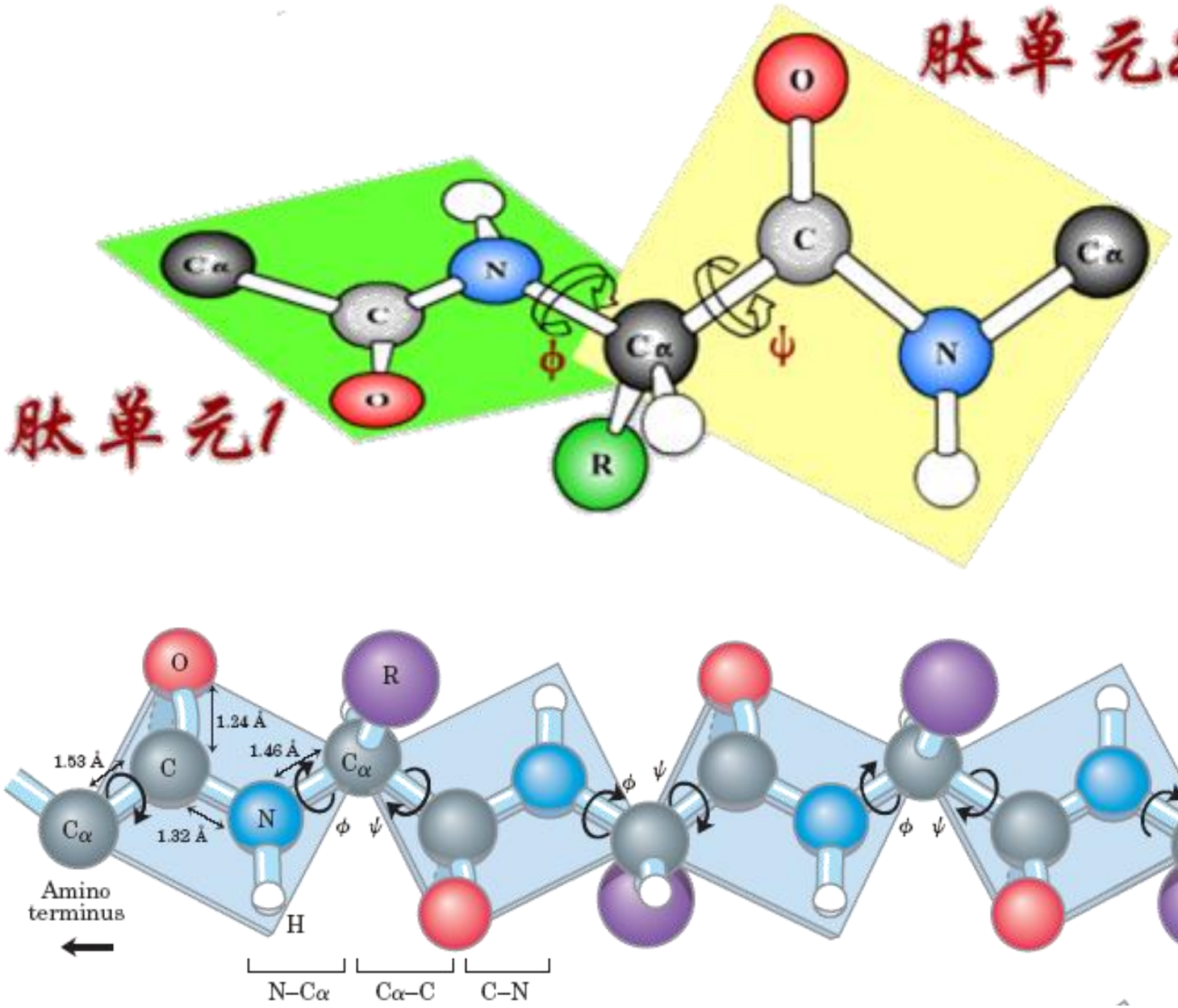
➢参与组成肽键的$C\alpha_1$、C、O、N、H、$C\alpha_2$ 6个原子在同一平面,称为肽键平面。

➢ **肽键反式构型**

➢ 0.127 nm ＜ 肽键键长 ＜0.149 nm

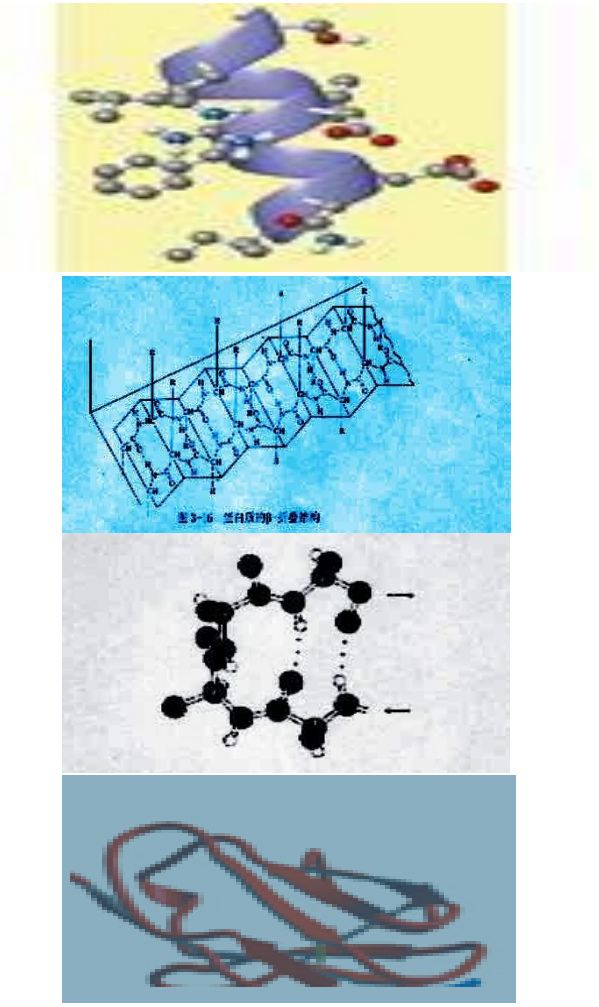
0.132nm

部分双键性质,不能自由旋转。


肽单元2
肽单元1
O
C
Cα
N
Cα
C
O
R
ϕ
ψ
O
R
1.24 Å
1.53 Å
1.46 Å
C
Cα
Cα
N
1.32 Å
ϕ
ψ
Amino terminus
H
Carboxyl terminus
N–Cα
Cα–C
C–N

蛋白质二级结构（满足主链氢键）的主要形式：

- α-螺旋 (α -helix)
- β-折叠 (β-pleated sheet)
- β-转角 (β-turn)
- 无规卷曲 (random coil)



1、α-螺旋 (α -helix)

α-螺旋结构要点：

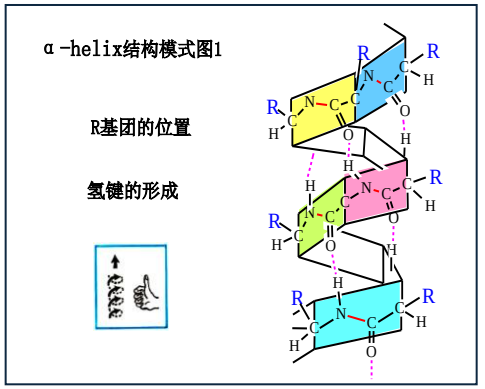
①外观：右手螺旋（顺）

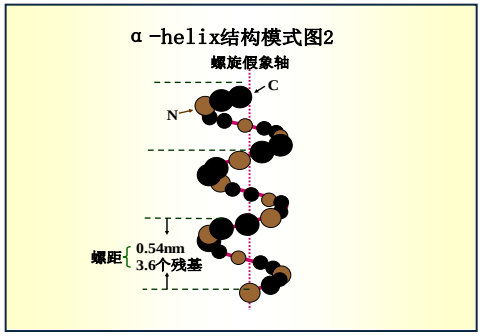
②螺距：3.6个氨基酸、0.54nm

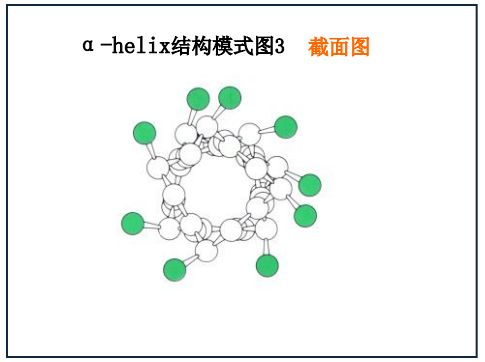
③稳定因素：氢键

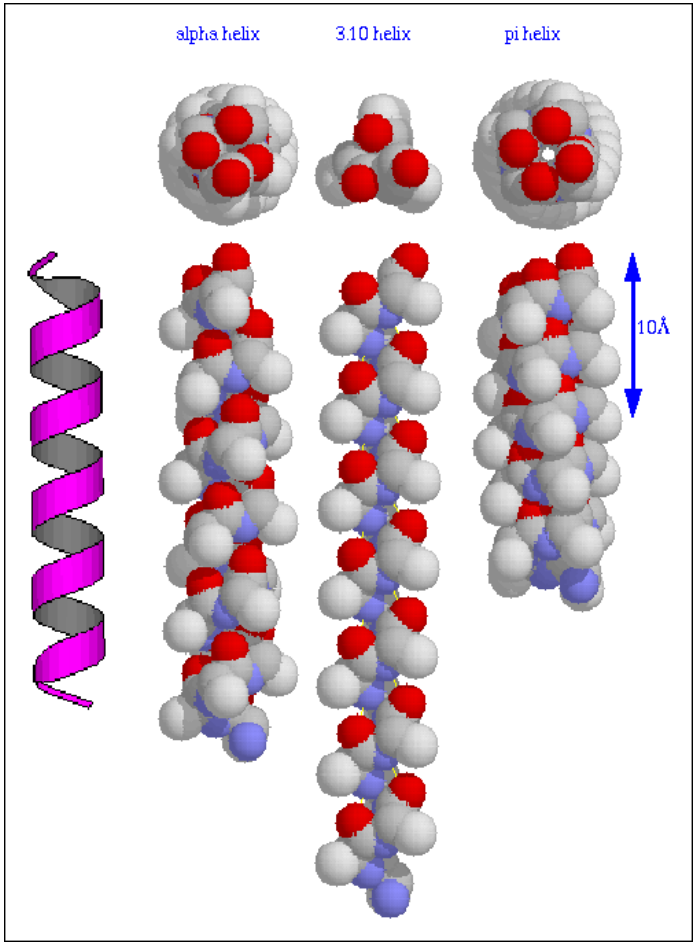
④R基团的位置：螺旋外侧，

R基团影响α-螺旋的形成

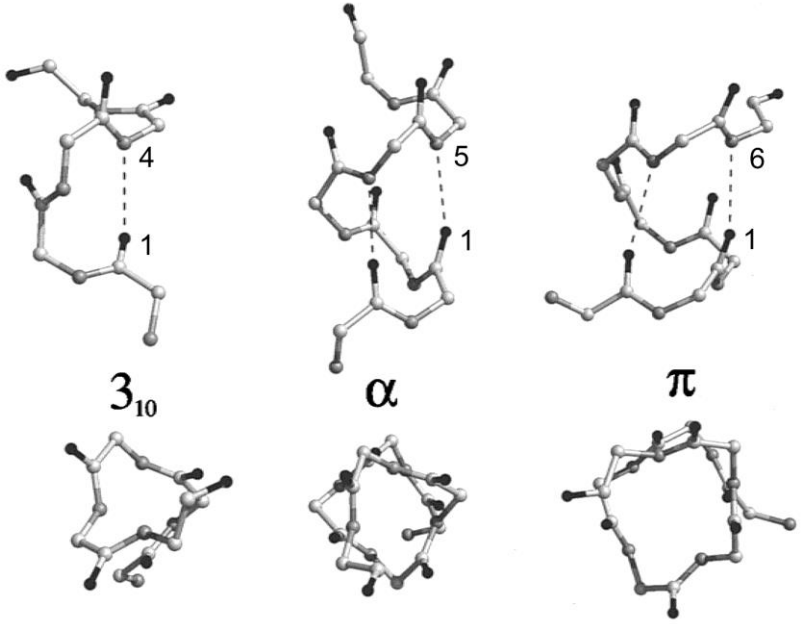








α, 3_{10} and π-helices



helix type	frequency	phi (ϕ)	psi (ψ)	residues per turn	rise (Å) per residue	pitch (Å)	H-bond (CO,NH)	atoms in H-bonded loop	radius (Å) (backbone)
α	abundant	-57.8	-47.0	3.6	1.5	5.5	i, i+4	13	2.3
3_{10}	infrequent	-74.0	-4.0	3.0	2.0	6.0	i, i+3	10	1.9
π	rare	-57.1	-69.7	4.4	1.1	5.0	i, i+5	16	2.8
β-strand	abundant	-139.0	135.0	2.0	3.4	6.8			

Right handed α-helix

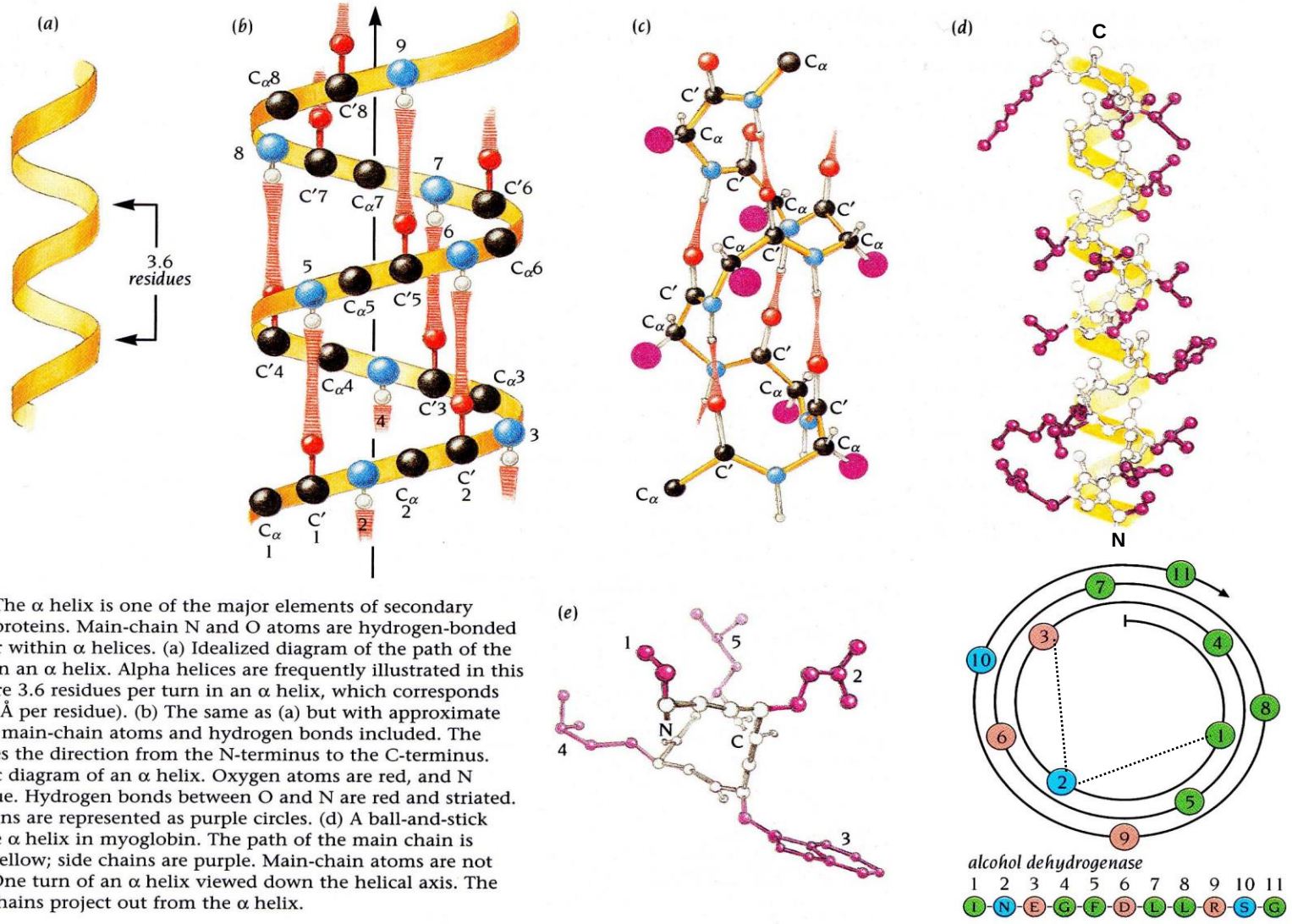


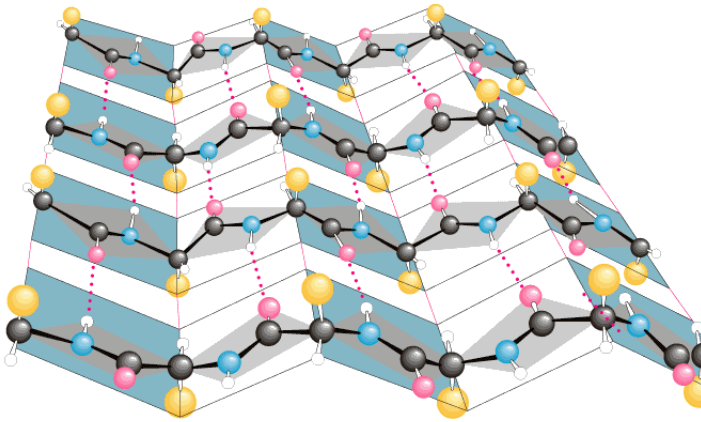
Figure 2.2 The α helix is one of the major elements of secondary structure in proteins. Main-chain N and O atoms are hydrogen-bonded to each other within α helices. (a) Idealized diagram of the path of the main chain in an α helix. Alpha helices are frequently illustrated in this way. There are 3.6 residues per turn in an α helix, which corresponds to 5.4 Å (1.5 Å per residue). (b) The same as (a) but with approximate positions for main-chain atoms and hydrogen bonds included. The arrow denotes the direction from the N-terminus to the C-terminus. (c) Schematic diagram of an α helix. Oxygen atoms are red, and N atoms are blue. Hydrogen bonds between O and N are red and striated. The side chains are represented as purple circles. (d) A ball-and-stick model of one α helix in myoglobin. The path of the main chain is outlined in yellow; side chains are purple. Main-chain atoms are not colored. (e) One turn of an α helix viewed down the helical axis. The purple side chains project out from the α helix.

影响α-螺旋稳定的因素：

(1) 酸性或碱性AA集中的区域（同种电荷的相斥）；

(2) 脯氨酸，不利于α-螺旋形成（亚氨基酸）；

(3) 较大的R侧链集中，不利于α-螺旋形成
（空间位阻效应）。



2、β-折叠 (β-pleated sheet)

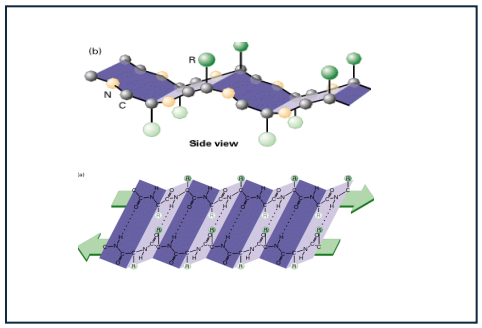
β-折叠结构要点：

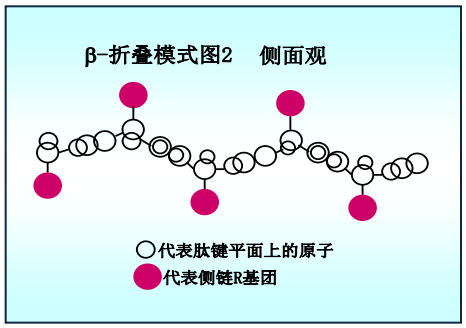
①肽段折纸状、锯齿状，相对伸展。

②几个肽段顺向平行、反向平行

③维持因素：氢键。

④R基团的位置：锯齿的上下方（外侧）。





Anti-parallel β-sheet

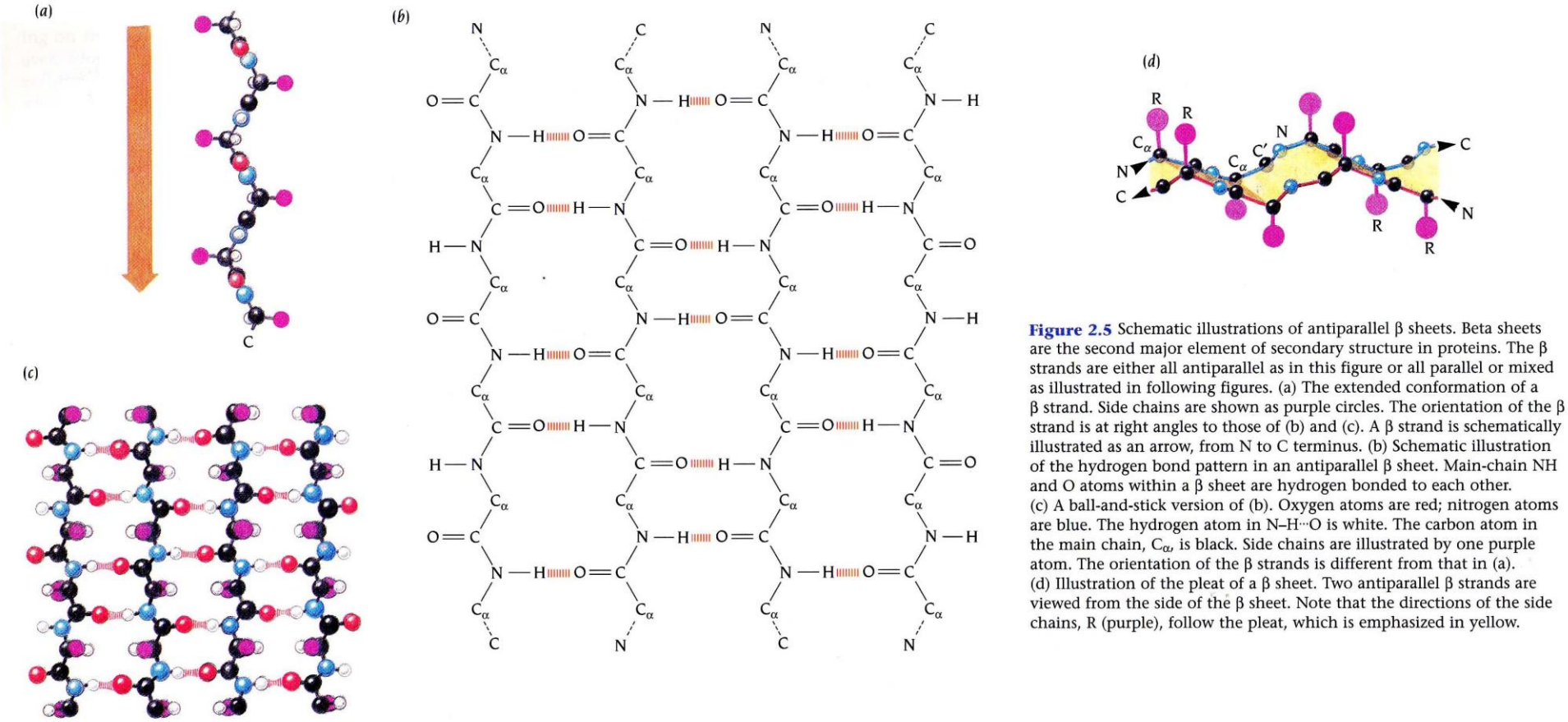


Figure 2.5 Schematic illustrations of antiparallel β sheets. Beta sheets are the second major element of secondary structure in proteins. The β strands are either all antiparallel as in this figure or all parallel or mixed as illustrated in following figures. (a) The extended conformation of a β strand. Side chains are shown as purple circles. The orientation of the β strand is at right angles to those of (b) and (c). A β strand is schematically illustrated as an arrow, from N to C terminus. (b) Schematic illustration of the hydrogen bond pattern in an antiparallel β sheet. Main-chain NH and O atoms within a β sheet are hydrogen bonded to each other. (c) A ball-and-stick version of (b). Oxygen atoms are red; nitrogen atoms are blue. The hydrogen atom in N–H···O is white. The carbon atom in the main chain, C_α, is black. Side chains are illustrated by one purple atom. The orientation of the β strands is different from that in (a). (d) Illustration of the pleat of a β sheet. Two antiparallel β strands are viewed from the side of the β sheet. Note that the directions of the side chains, R (purple), follow the pleat, which is emphasized in yellow.

Parallel β-sheet

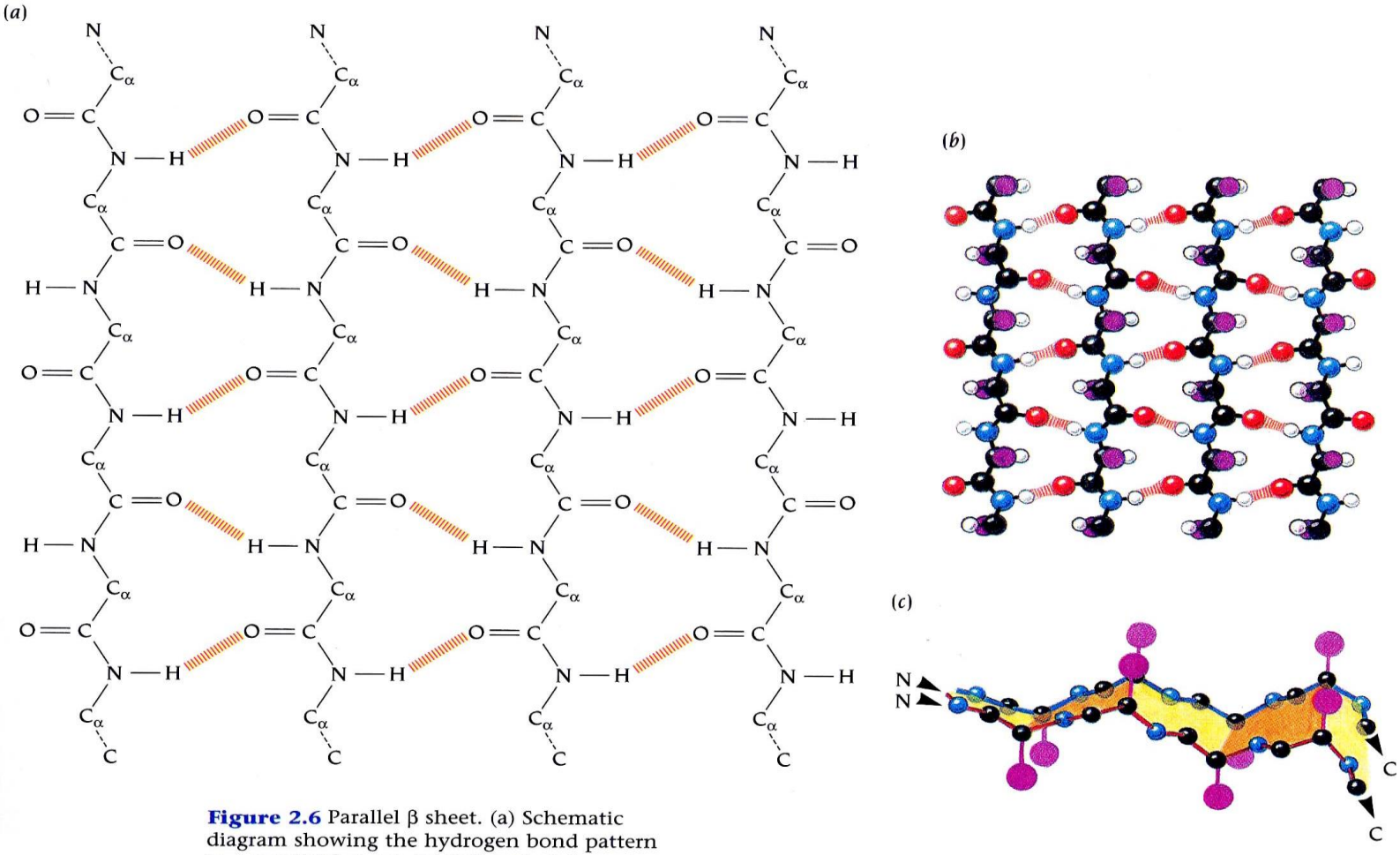
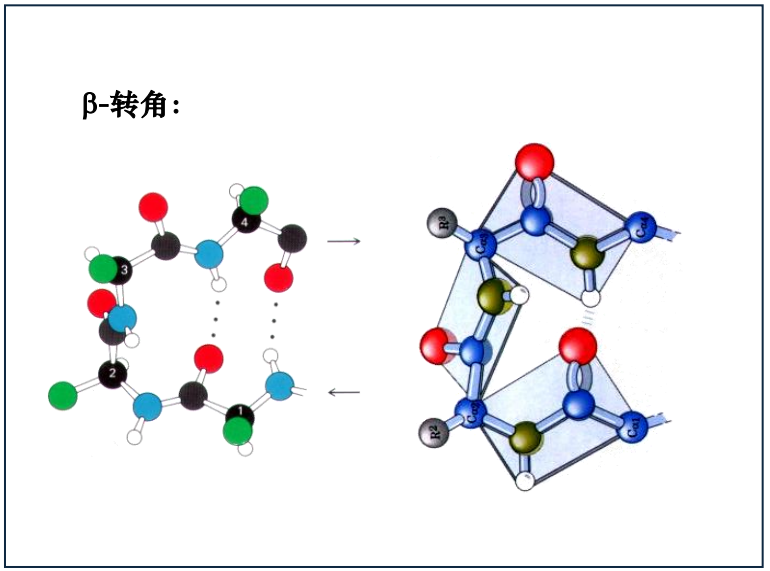


Figure 2.6 Parallel β sheet. (a) Schematic diagram showing the hydrogen bond pattern in a parallel β sheet. (b) Ball-and-stick version of (a). The same color scheme is used as in Figure 2.5c. (c) Schematic diagram illustrating the pleat of a parallel β sheet.

3. β-转角

- 4个氨基酸残基
- 第二个常为脯氨酸
- 180°回折
- 氢键维持

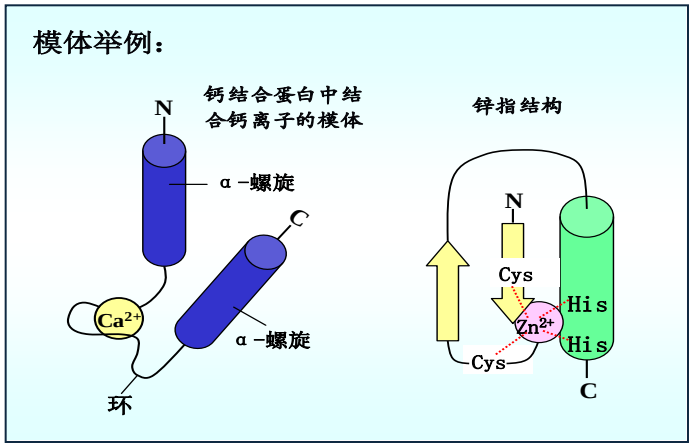


4. 无规卷曲

无规卷曲是用来阐述没有确定规律性的那部分肽链结构。

（五）模体（motif）或超二级结构

- 举例

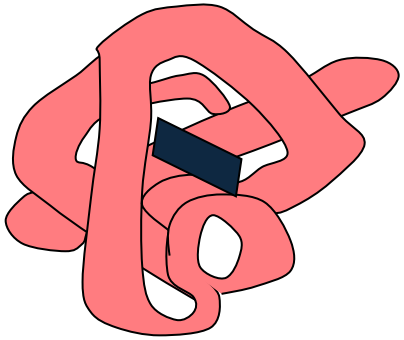


- 定义
  - 二个或二个以上二级结构
  - 空间上靠近
  - 特殊的构象
  - 特殊的功能

（六）氨基酸残基的侧链对二级结构形成的影响

蛋白质二级结构是以一级结构为基础的。一段肽链其氨基酸残基的侧链适合形成α-螺旋或β-折叠，它就会出现相应的二级结构。

三、蛋白质的三级结构(tertiary structure)

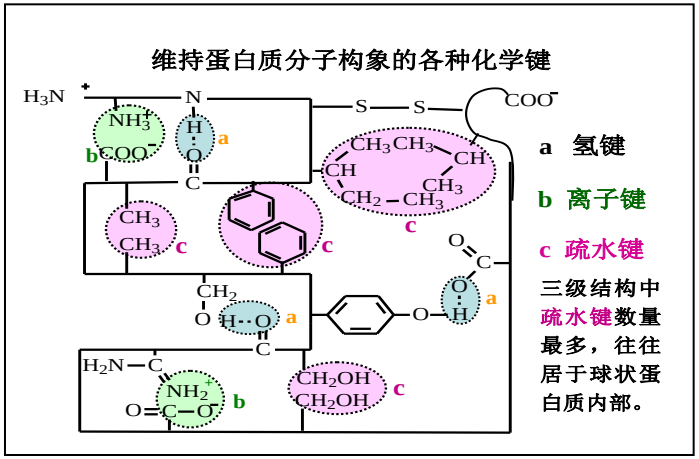


（一）定义：

{ 整条肽链
  全部氨基酸
  所有原子 } 三维空间的位置排布

主要靠非共价键：疏水键、离子键、氢键、范德华力等。

此外还有共价键：二硫键



Stable structure:

1. **Minimal free energy:**

   $\Delta G = \Delta H - T \Delta S$

2. **Alike stays with alike:**

   Bury sufficient hydrophobic groups;
   Not create (too) many internal cavities;
   Satisfy (most) hydrogen bond donors and acceptors;
   Pamper the buried charges;
   Avoid (most) side chain and main chain strain.

Ridge formed by side chains in an α-helix

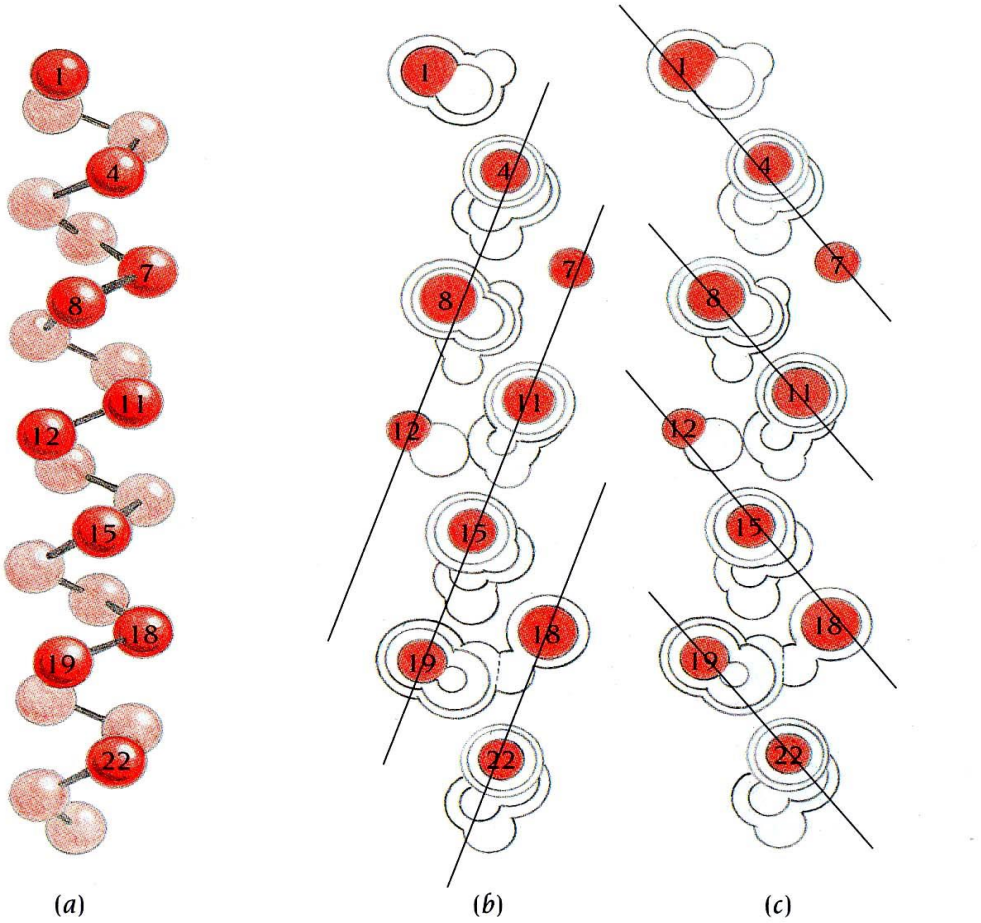


Figure 3.11 The side chains on the surface of an α helix form ridges separated by grooves, as schematically illustrated here. (a) An α helix with each residue represented by the first atom in the side chain, C_β. (b) The surface relief of a polyalanine α helix in the orientation shown in (a). Sections are cut through a space-filling model and superimposed. The residue numbers are placed on the side-chain atom. The ridges caused by the side chains separated by four residues are shown as lines. (c) The same as (b), but here the ridges are caused by side chains separated by three residues.

Side chain ridges determine helix packing

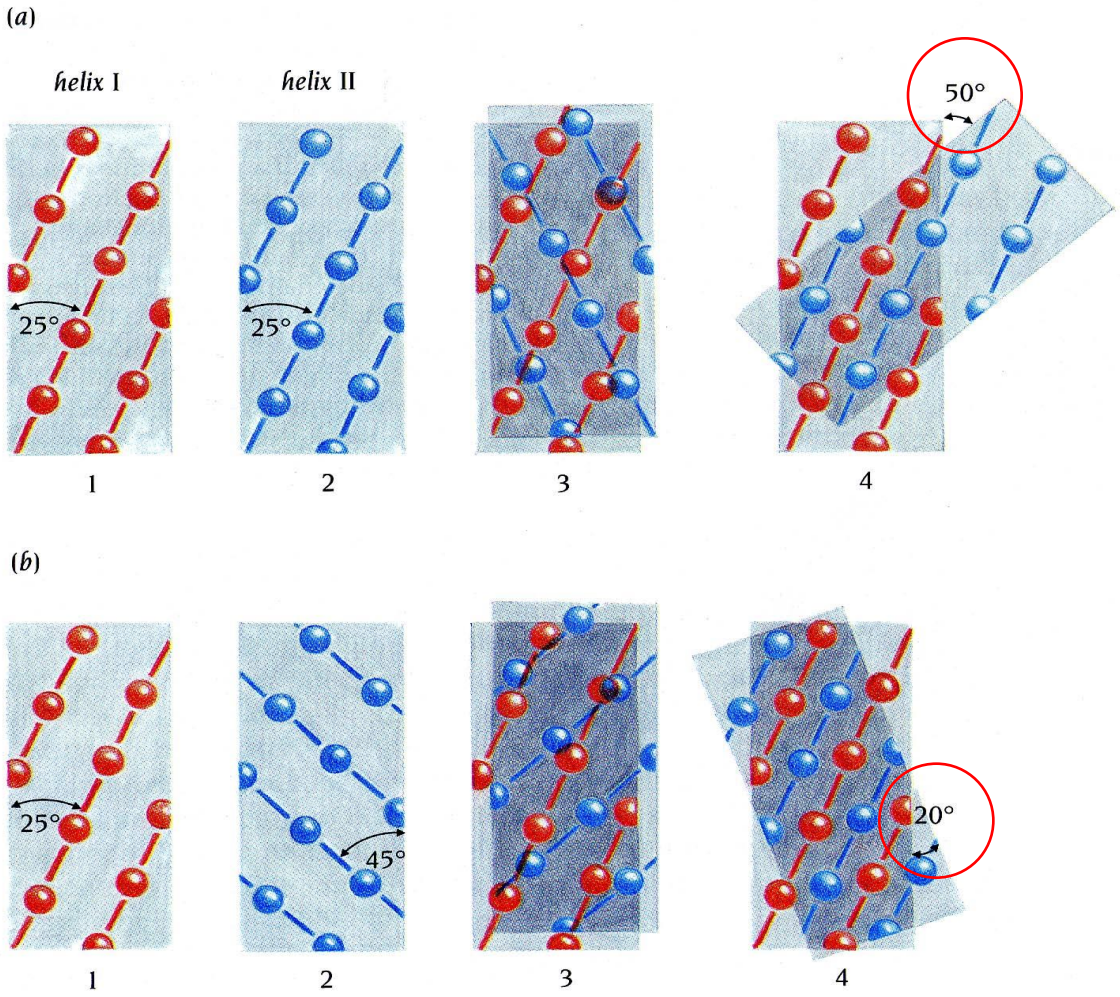


Figure 3.12 By fitting the ridges of side chains from one helix into the grooves between side chains of the other helix and vice versa, α helices pack against each other. (a) Two α helices, I and II, with ridges from side chains separated by four residues marked in red and blue, respectively. Panels 1 and 2 are the same view of the two α helices. In panel 3 the blue α helix is turned over through 180° in order to form an interface with the red α helix. In panel 4 the orientation of the helices has been rotated 50° in order to pack the ridges of one α helix into the grooves of the other. (b) In the red α helix the ridges are formed by side chains separated by four residues and in the blue α helix by three residues. The α helices are rotated 20° in order to pack ridges into grooves, in a direction opposite that in (a). (Adapted from C. Chothia et al., *Proc. Natl. Acad. Sci. USA* 74: 4130–4134, 1977.)

Structure of the c-Jun/c-Fos Leucine Zipper

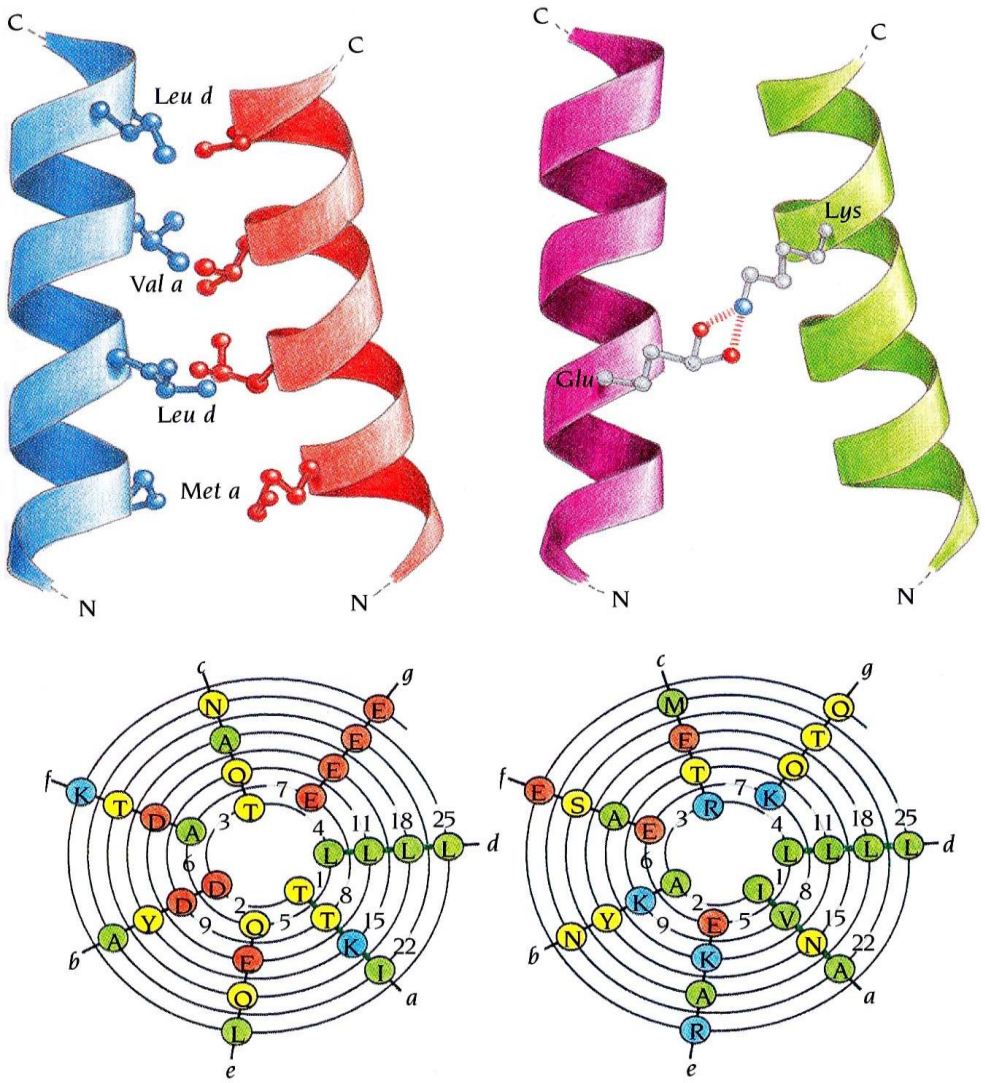


Figure 10.18 Side-chain interactions in leucine zipper structure. (a) The hydroph side chains in spikes *a* and *d* (see Figure form a hydrophobic core between the tw coiled α helices. (b) Charged side chains spikes *e* and *g* can promote dimer forma forming complementary charge interacti between the two α helices.

Four helix bundle

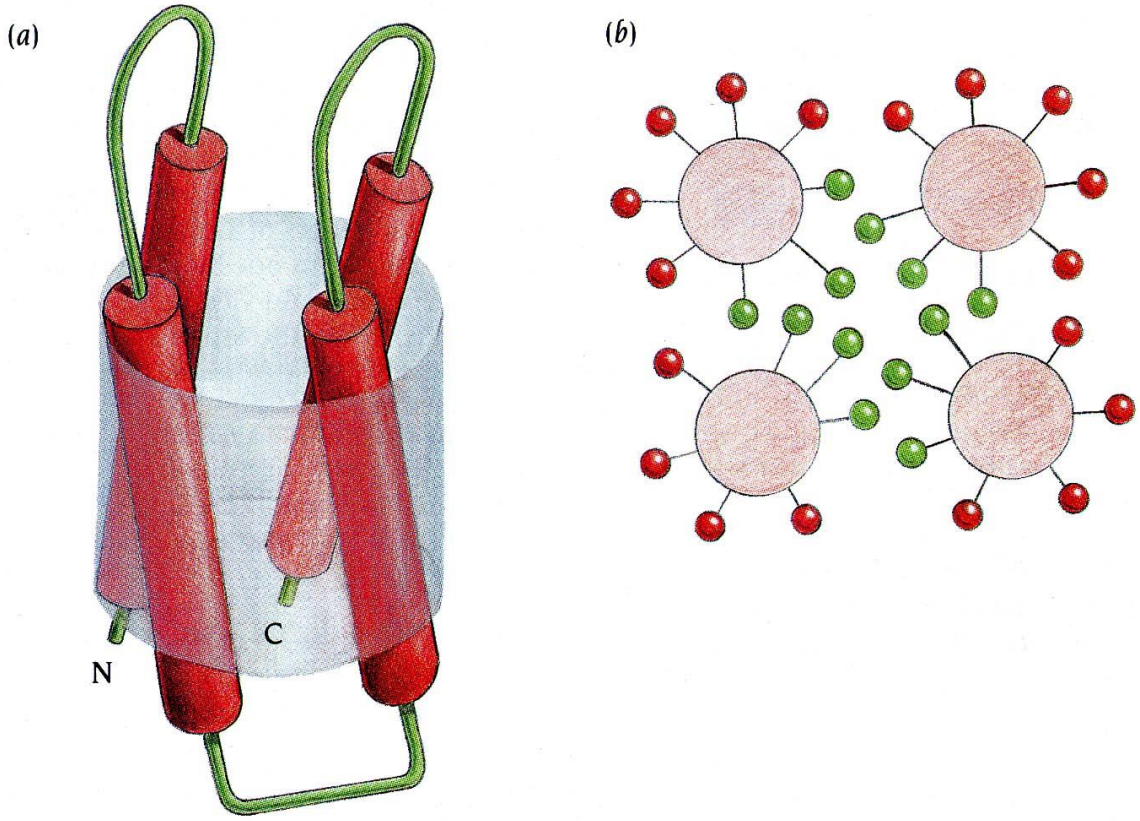
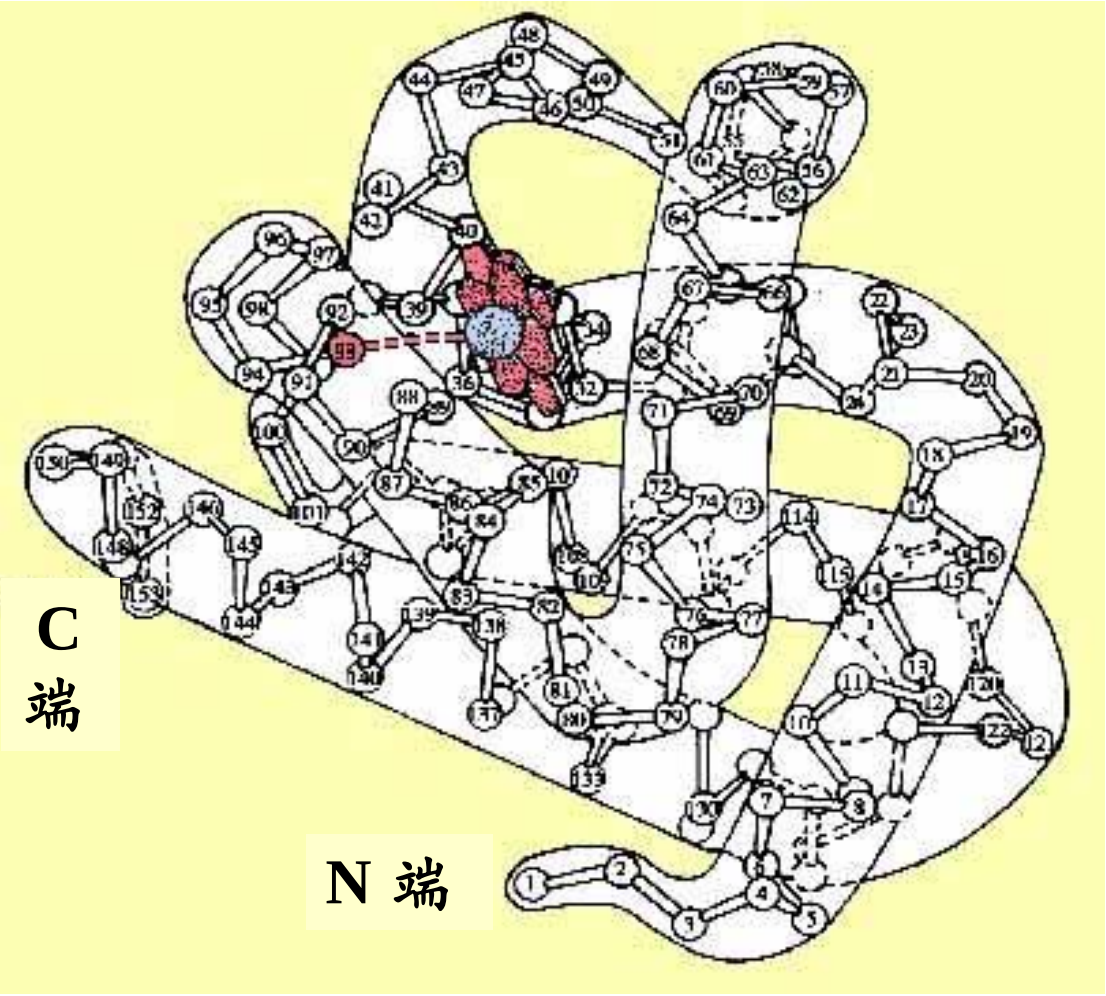
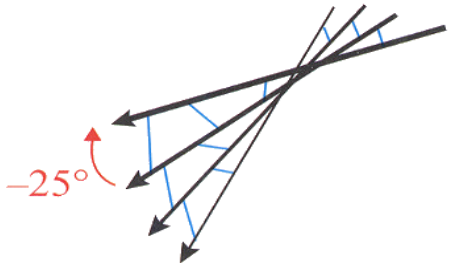


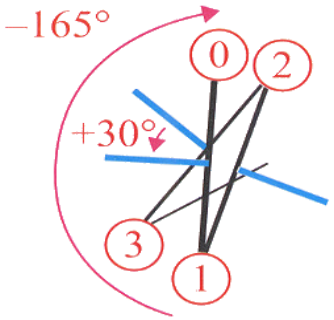
Figure 3.6 Four-helix bundles frequently occur as domains in α proteins. The arrangement of the α helices is such that adjacent helices in the amino acid sequence are also adjacent in the three-dimensional structure. Some side chains from all four helices are buried in the middle of the bundle, where they form a hydrophobic core. (a) Schematic representation of the path of the polypeptide chain in a four-helix-bundle domain. Red cylinders are α helices. (b) Schematic view of a projection down the bundle axis. Large circles represent the main chain of the α helices; small circles are side chains. Green circles are the buried hydrophobic side chains; red circles are side chains that are exposed on the surface of the bundle, which are mainly hydrophilic. [(a) Adapted from P.C. Weber and F.R. Salemme, *Nature* 287: 82–84, 1980.]

- 肌红蛋白 (Mb)

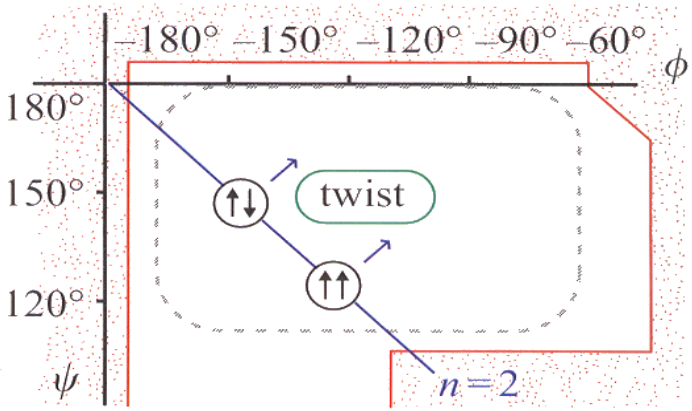




(a)



(b)



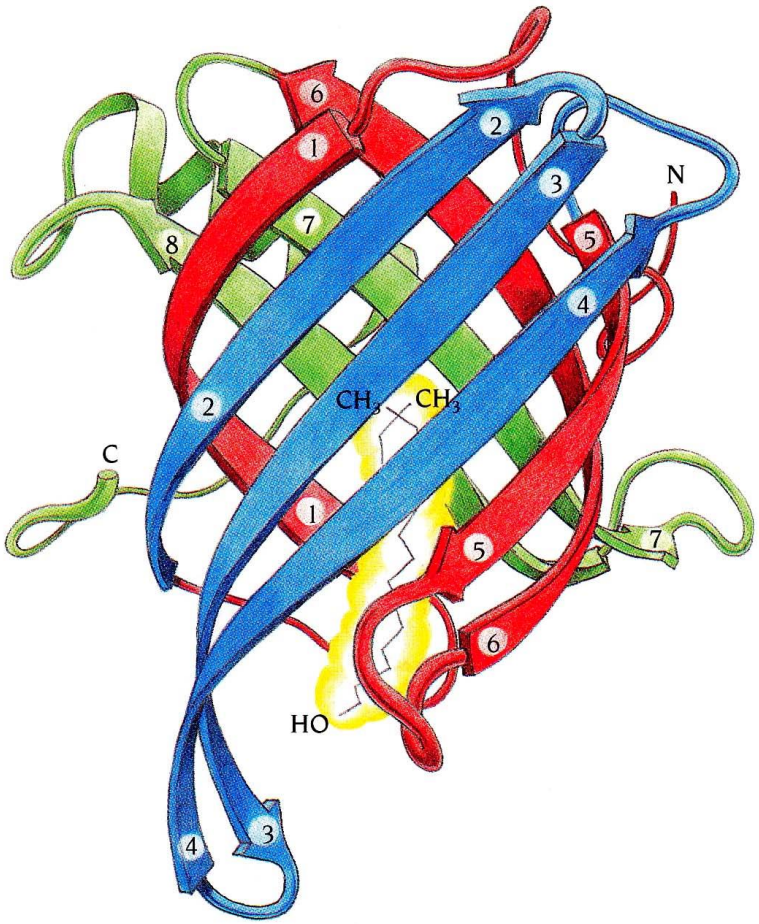
(c)

Almost all beta-sheets have a left-handed twist

(when looked perpendicular to the beta-strands in the plane of the sheet)

(From Finkelstein and Ptitsyn - "Protein Physics")

β-barrel structure



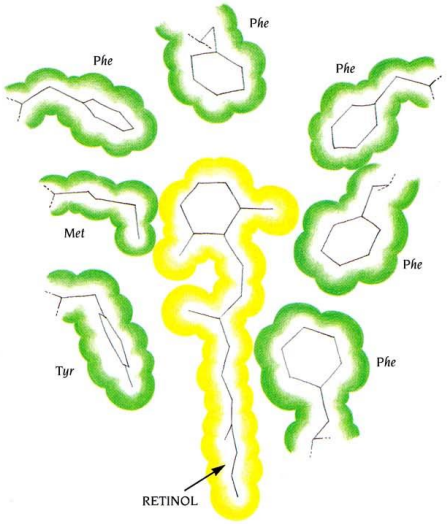
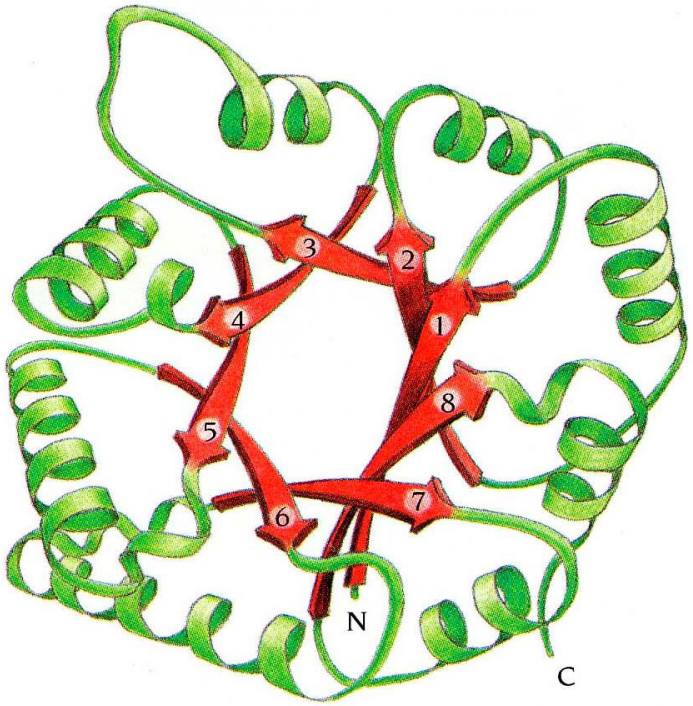
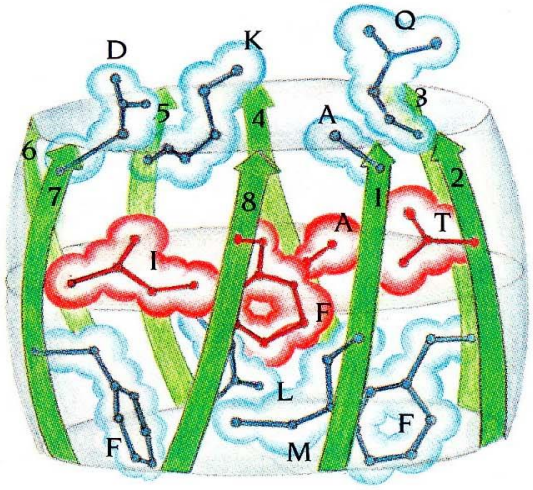
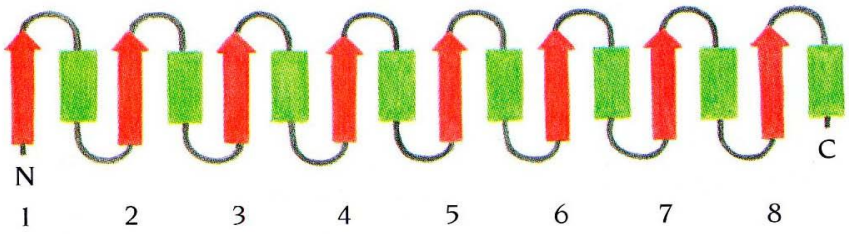


Figure 5.3 Schematic diagram of the structure of human plasma retinol-binding protein (RBP), which is an up-and-down β barrel. The eight antiparallel β strands twist and curl such that the structure can also be regarded as two β sheets (green and blue) packed against each other. Some of the twisted β strands (red) participate in both β sheets. A retinol molecule, vitamin A (yellow), is bound inside the barrel, between the two β sheets, such that its only hydrophilic part (an OH tail) is at the surface of the molecule. The topological diagram of this structure is the same as that in Figure 5.2. (Courtesy of Alwyn Jones, Uppsala, Sweden.)

TIM barrel (α/β)

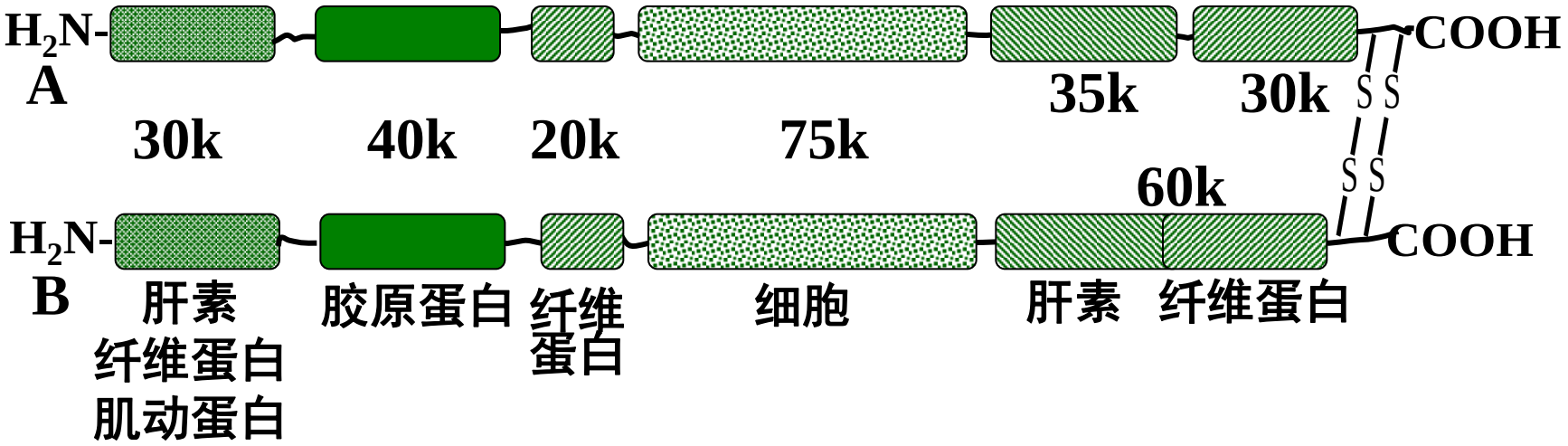






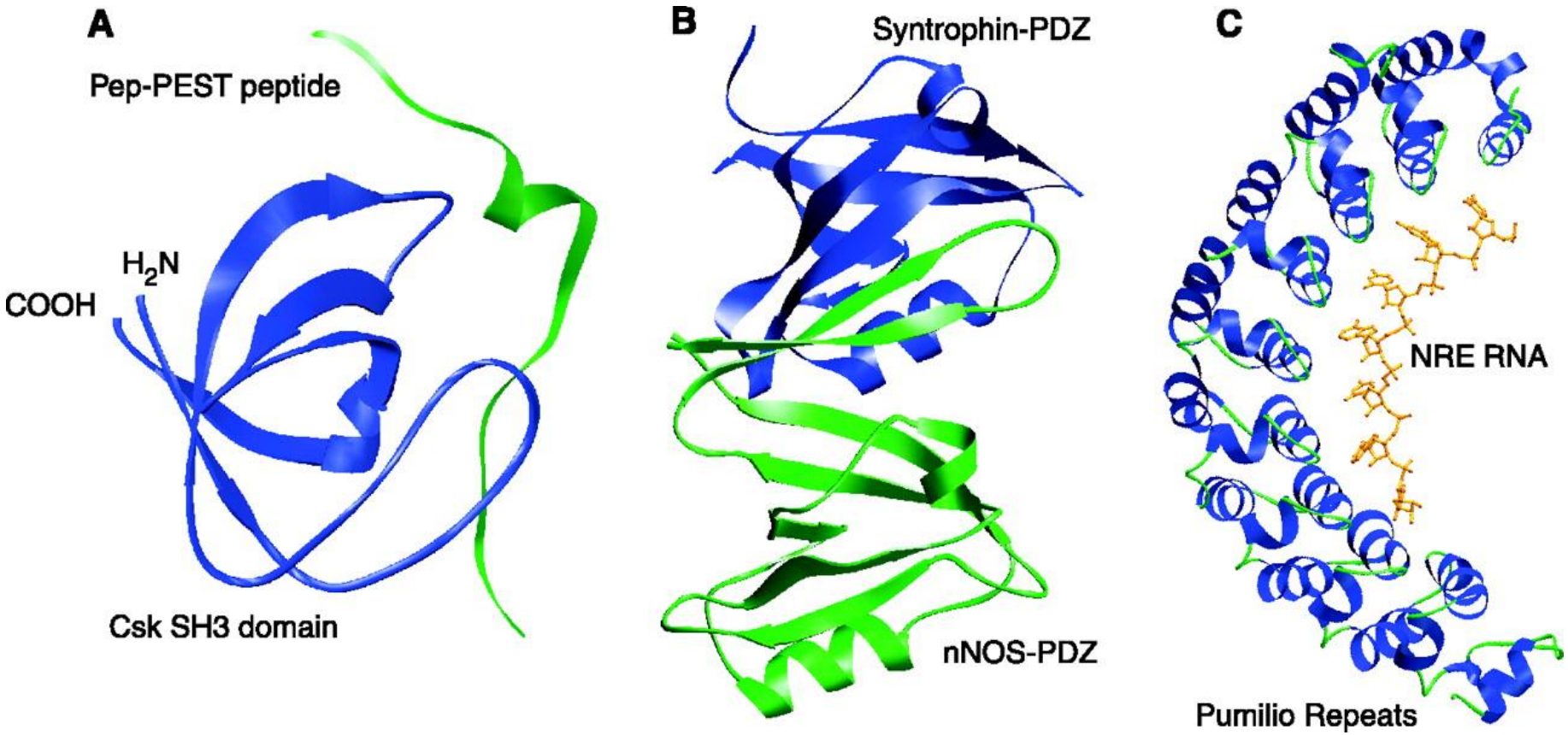
（二）结构域(domain)

- 举例　纤连蛋白(fibronectin)分子的结构域



- **结构域：**三级结构中分割成折叠较为紧密的区域，各行使其功能。

Conserved domains: functional units preserved during evolution



（三）分子伴侣(chaperon)

分子伴侣通过提供一个保护环境从而加速蛋白质折叠成天然构象或形成四级结构。

具体作用：

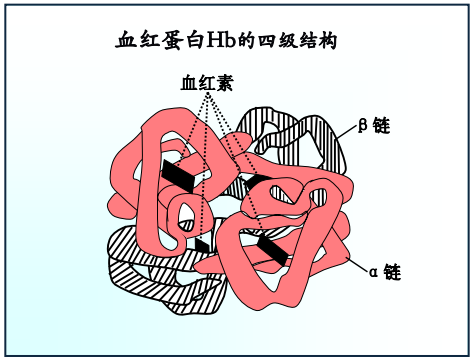
1.能可逆地与未折叠肽段的疏水部分结合随后松开，如此重复进行可防止错误的聚集发生，使肽链正确折叠。

2.可与错误聚集的肽段结合，使之解聚后，再诱导其正确折叠。

3. 在蛋白质分子折叠过程中二硫键的正确形成起了重要的作用。

四、蛋白质的四级结构(quaternary structure)

•**四级结构**— 2或2条以上多肽链，以非共价键相连，链与链之间的空间布局。



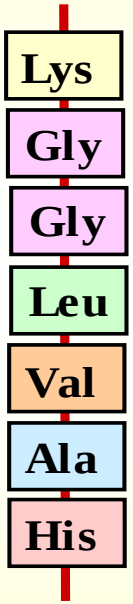
•**亚基 (subunit)**— 四级结构中每条具有完整三级结构的多肽链

•**维持因素**— 次级键：氢键、离子键
无共价键

蛋白质结构层次比较

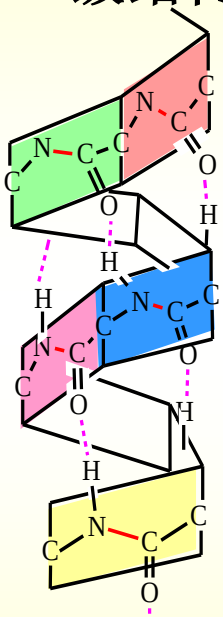
蛋白质结构层次的比较

一级结构



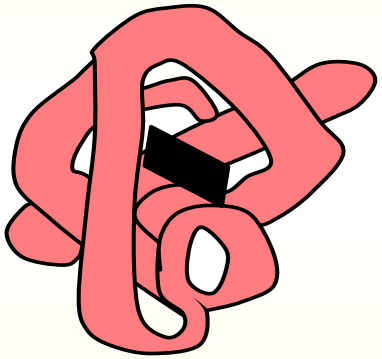
氨基酸序列

二级结构



主链原子局部构象

三级结构



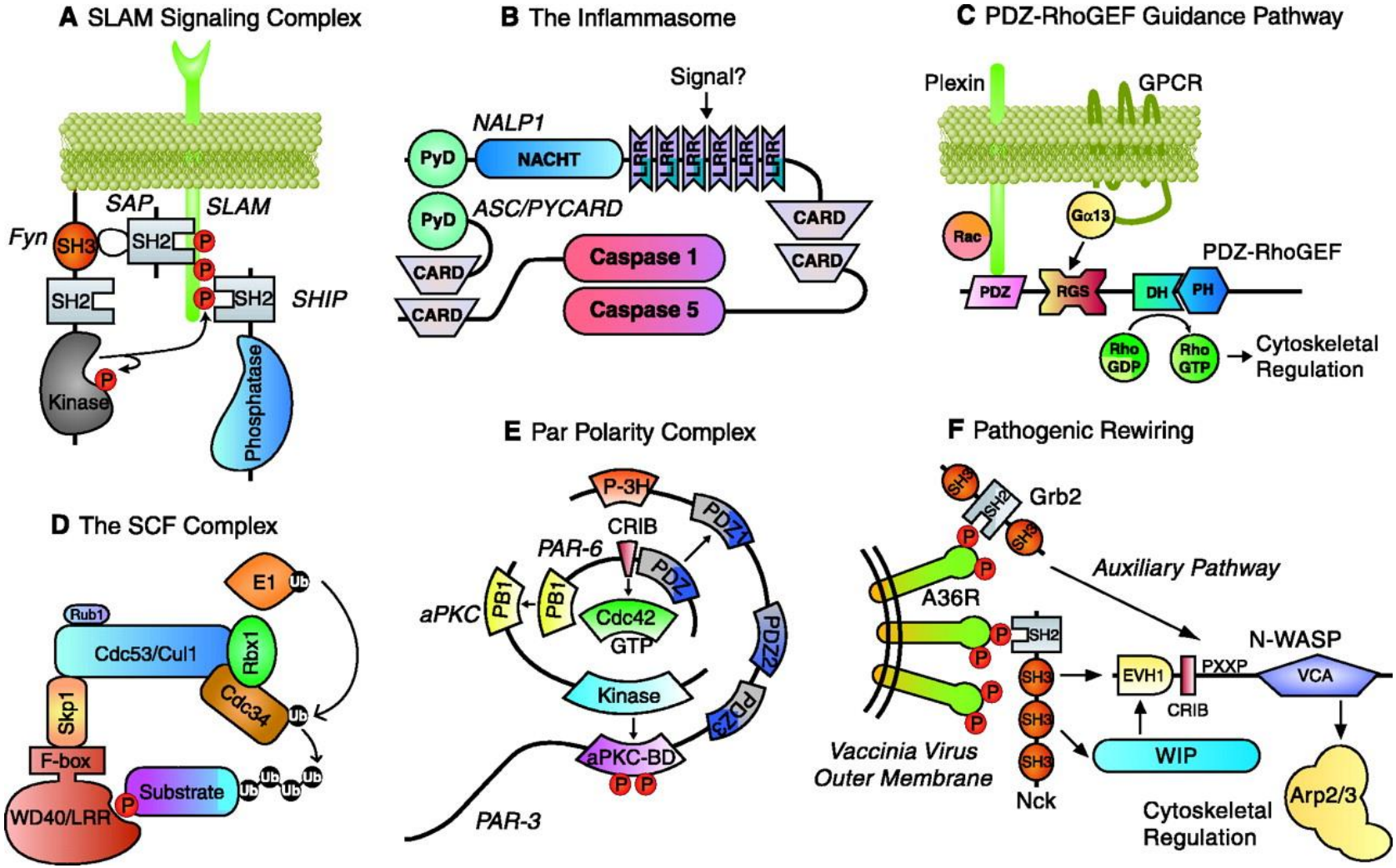
一条链所有原子的位置

四级结构



亚基间的相对位置

多层次结构间的能量调和
及亚稳态对可调控系统的重要性



第三节

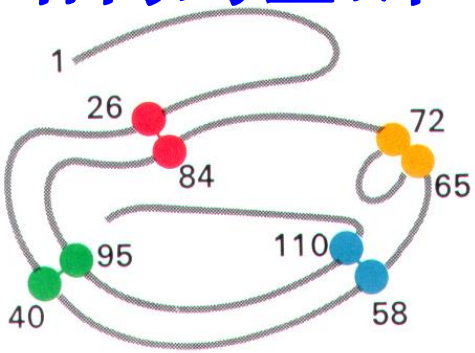
蛋白质结构与功能的关系

The Relation of Structure and Function of Protein

一级结构决定空间结构
空间结构决定生物学功能

一、蛋白质一级结构与功能的关系

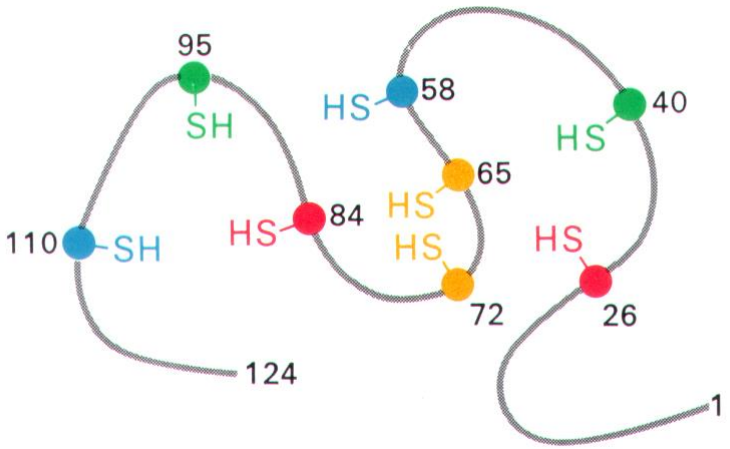
（一）一级结构是空间结构的基础



天然状态，
有催化活性

去除尿素、
β-巯基乙醇

尿素、β-巯基乙醇



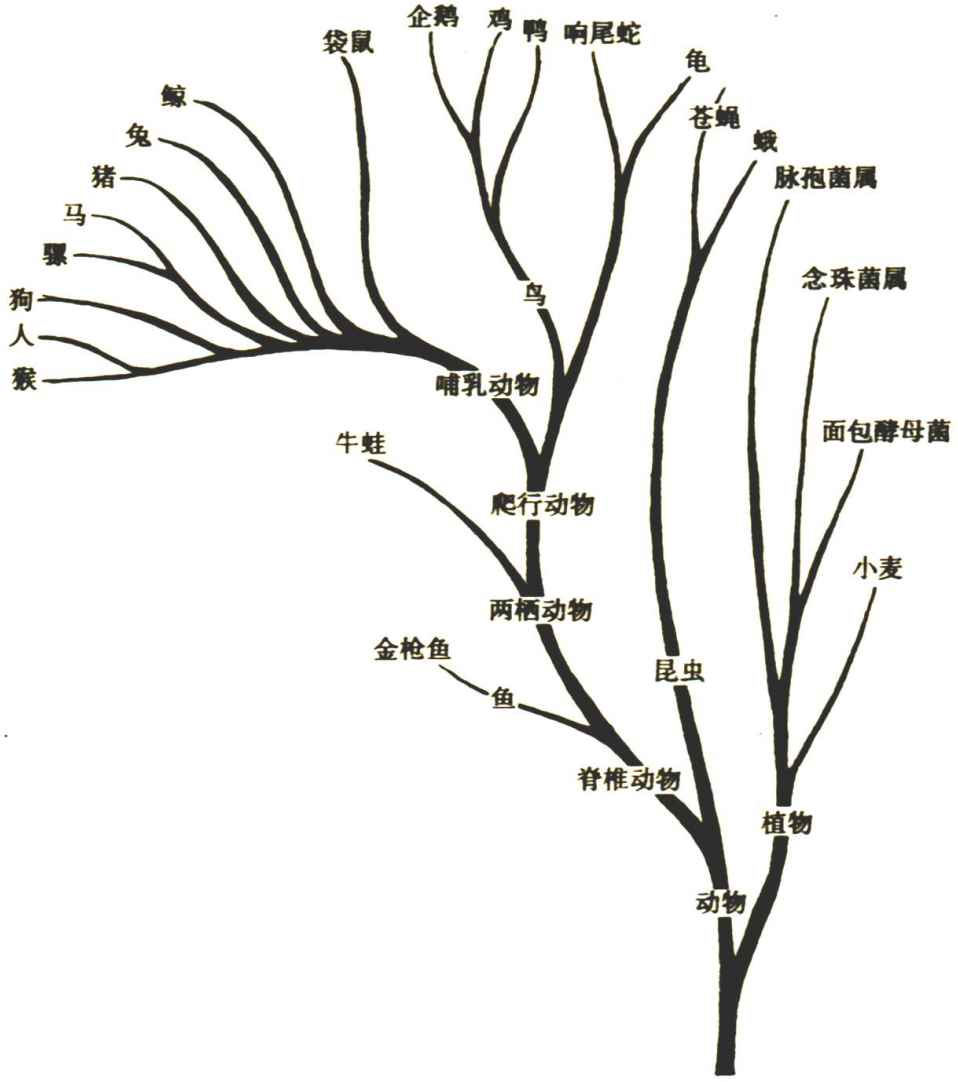
非折叠状态，无活性

（二）相似结构具有相似的高级结构与功能

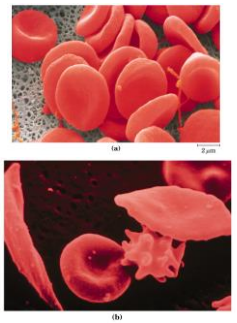
胰岛素	氨基酸残基序号			
	A5	A6	A10	B30
人	Thr	Ser	Ile	Thr
猪	Thr	Ser	Ile	Ala
狗	Thr	Ser	Ile	Ala
兔	Thr	Gly	Ile	Ser
牛	Ala	Gly	Val	Ala
羊	Ala	Ser	Val	Ala
马	Thr	Ser	Ile	Ala

（三）氨基酸序列提供重要的生物化学信息

一些广泛存在于生物界的蛋白质如细胞色素C，比较它们的一级结构，可以帮助了解物种进化间的关系。



（四）关键结构改变可引起疾病



■例：镰刀形红细胞贫血

HbA β 肽 链

N-val · his · leu · thr · pro · glu · glu · · · · ·C(146)

HbS β 肽链

N-val · his · leu · thr · pro · val · glu · · · · ·C(146)

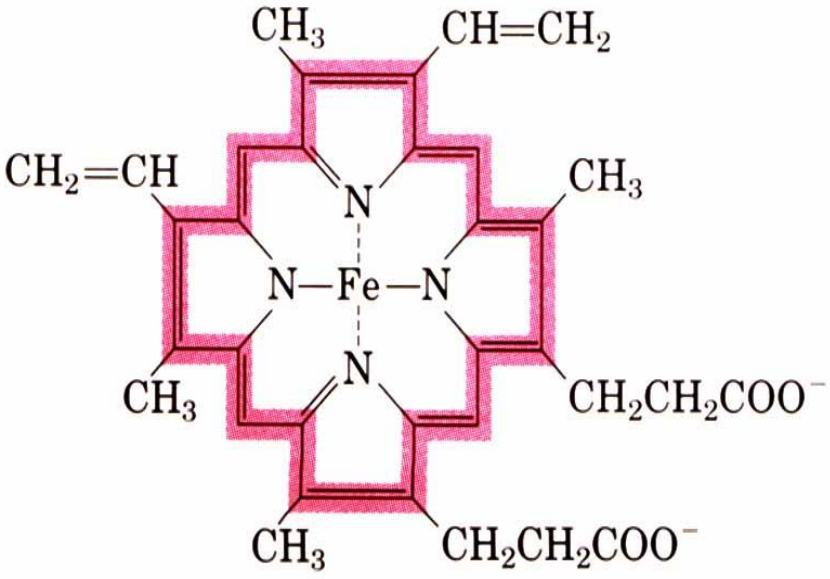
这种由基因突变引起蛋白质分子发生变异所导致的疾病，称为“分子病”。

二、蛋白质的功能依赖特定空间结构

（一）血红蛋白亚基与肌红蛋白结构相似

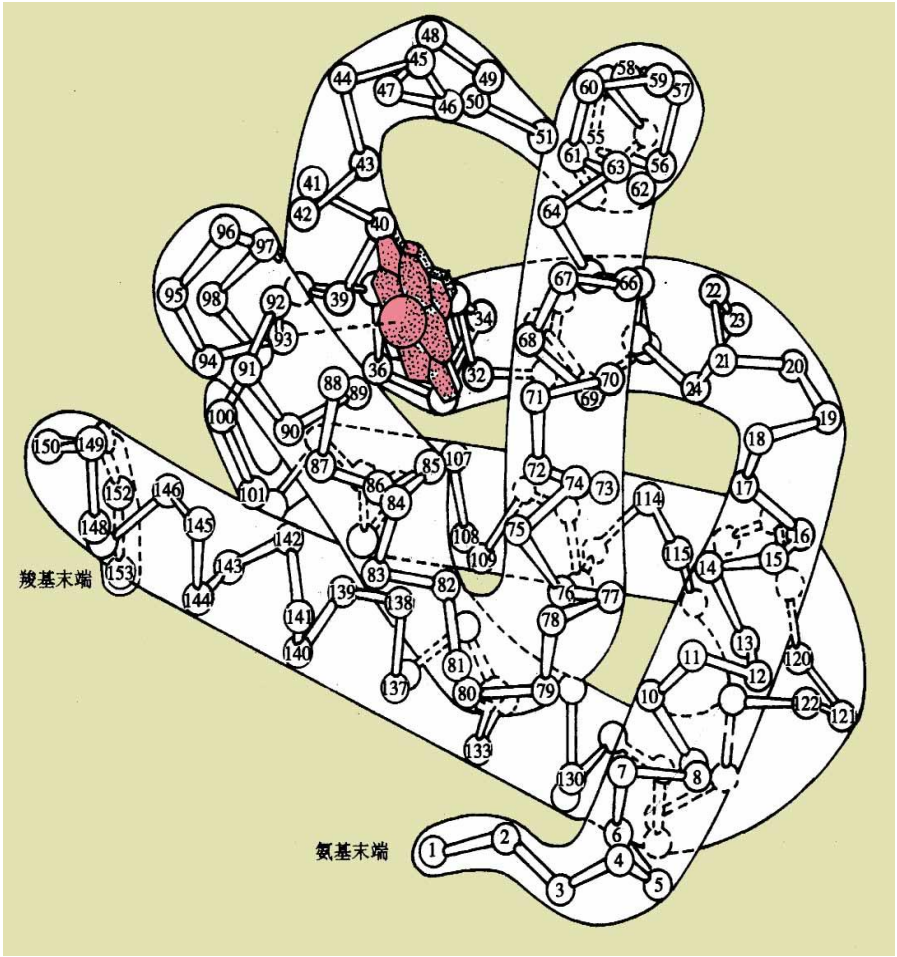
- 肌红蛋白/血红蛋白含有血红素辅基

血红素结构



Iron protoporphyrin IX
(in *b*-type cytochromes)

■肌红蛋白(myoglobin，Mb)



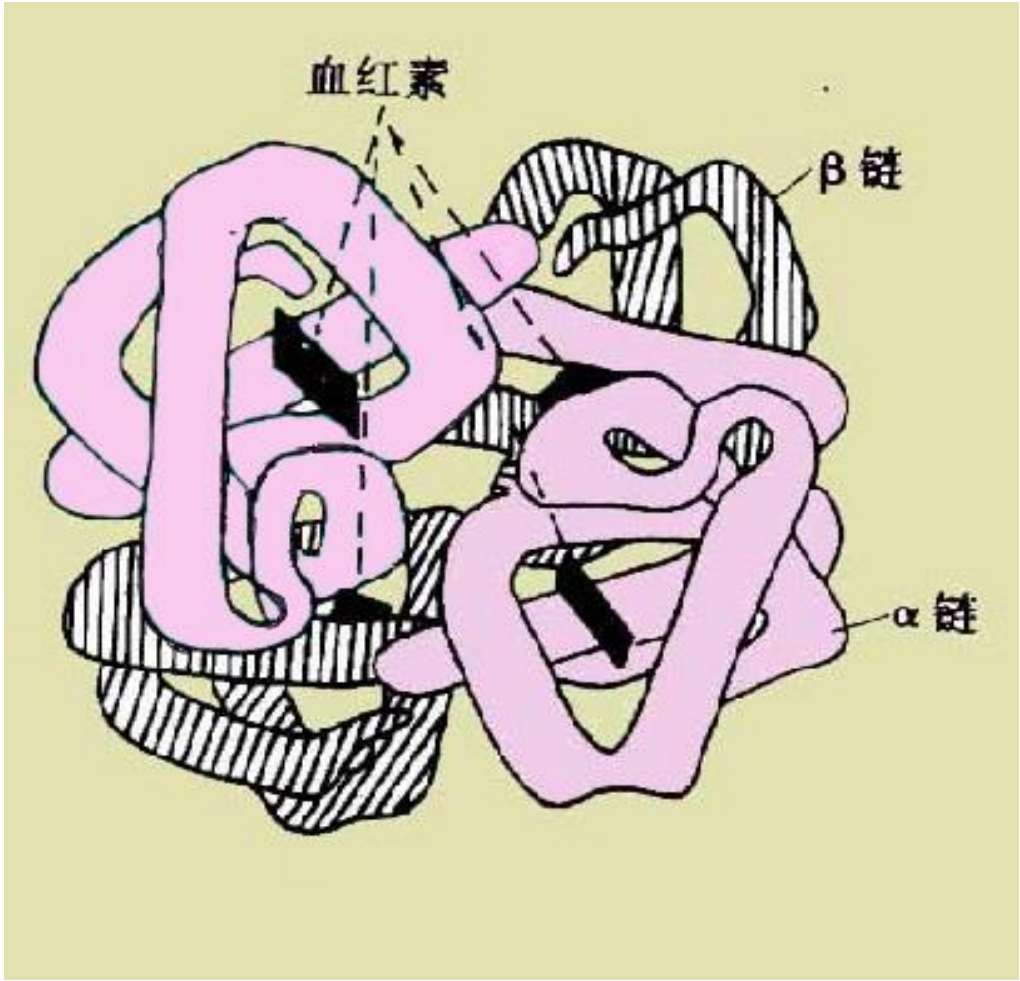
肌红蛋白是一个只有三级结构的单链蛋白质，有8段α-螺旋结构。

血红素分子中的两个丙酸侧链以离子键形式与肽链中的两个碱性氨基酸侧链上的正电荷相连，加之肽链中的F8组氨酸残基还与Fe^{2+}形成配位结合，所以血红素辅基与蛋白质部分稳定结合。

■血红蛋白(hemoglobin，Hb)

血红蛋白具有4个亚基组成的四级结构，每个亚基可结合1个血红素并携带1分子氧。

Hb亚基之间通过8对盐键，使4个亚基紧密结合而形成亲水的球状蛋白。



（二）血红蛋白亚基构象变化可影响亚基与氧结合

Hb与Mb一样能可逆地与O_2结合， Hb与O_2结合后称为氧合Hb。氧合Hb占总Hb的百分数（称百分饱和度）随O_2浓度变化而改变。

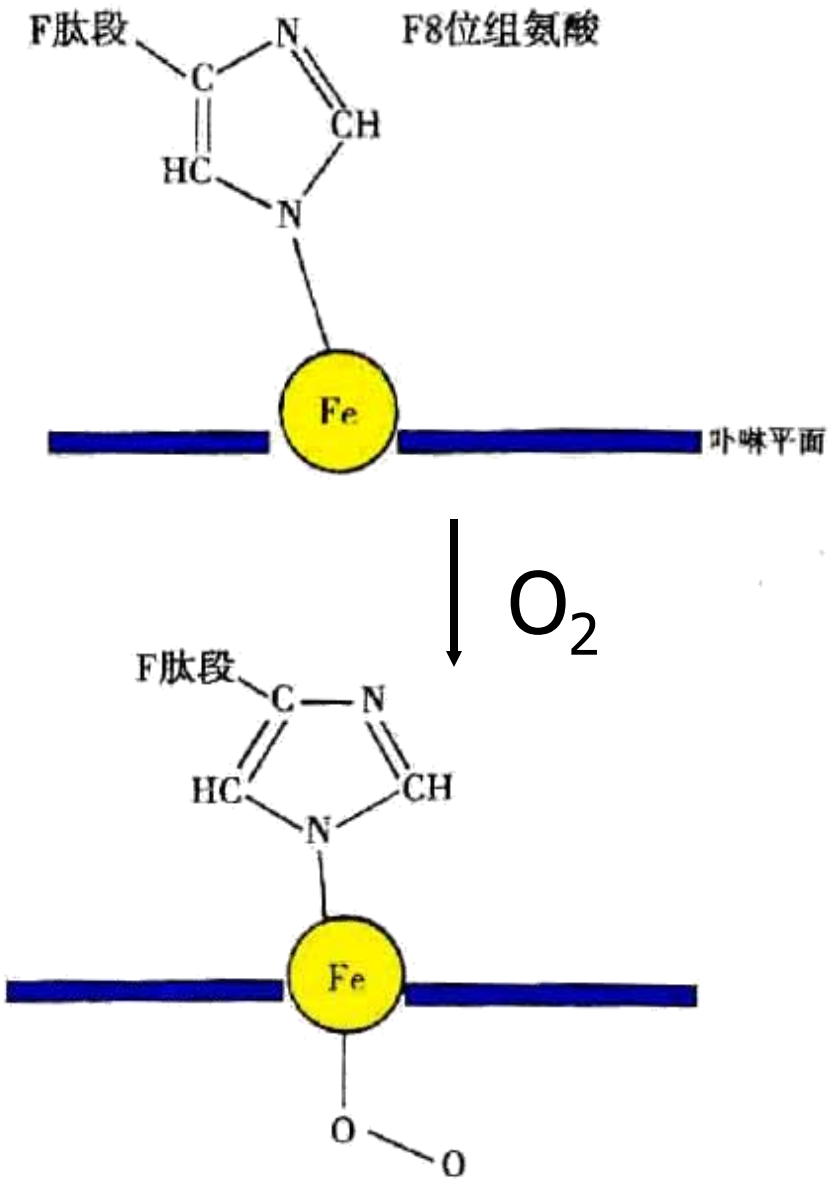
- 协同效应(cooperativity)

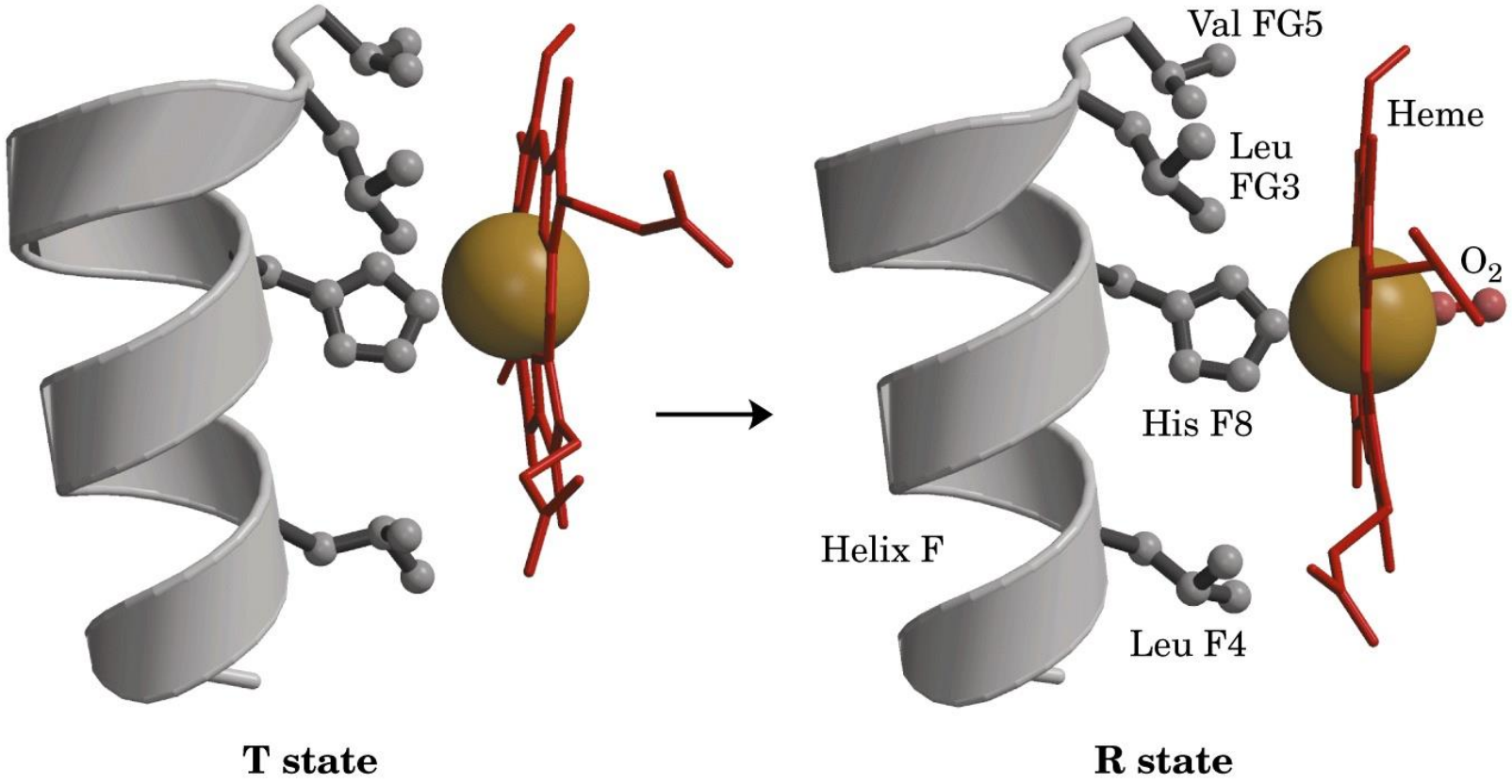
一个寡聚体蛋白质的一个亚基与其配体结合后，能影响此寡聚体中另一个亚基与配体结合能力的现象，称为协同效应。

如果是促进作用则称为正协同效应
(positive cooperativity)

如果是抑制作用则称为负协同效应
(negative cooperativity)

血红素与氧结合后，铁原子半径变小，就能进入卟啉环的小孔中，继而引起肽链位置的变动。




Val FG5
Heme
Leu
FG3
O_2
His F8
Helix F
Leu F4
T state
R state

■ 变构效应(allosteric effect)

蛋白质空间结构的改变伴随其功能的变化，称为变构效应。

（三）蛋白质构象改变可引起疾病

蛋白质构象疾病：若蛋白质的折叠发生错误，尽管其一级结构不变，但蛋白质的构象发生改变，仍可影响其功能，严重时可导致疾病发生。

蛋白质构象改变导致疾病的机理：有些蛋白质错误折叠后相互聚集，常形成抗蛋白水解酶的淀粉样纤维沉淀，产生毒性而致病，表现为蛋白质淀粉样纤维沉淀的病理改变。如:困扰欧洲经济和人们健康安全的疯牛病,就是一种叫朊病毒(prion)的蛋白质分子,研究发现了朊病毒通过一种可怕的拉链结构,会把蛋白质“拉链”内的水挤出来而脱水变成干燥的纤维状。蛋白质就在这种朊病毒的侵害下逐渐纤维化,原来的神经细胞组织也就形成了空泡和纤维相杂的结构,导致了患者大脑组织的海绵样病变。

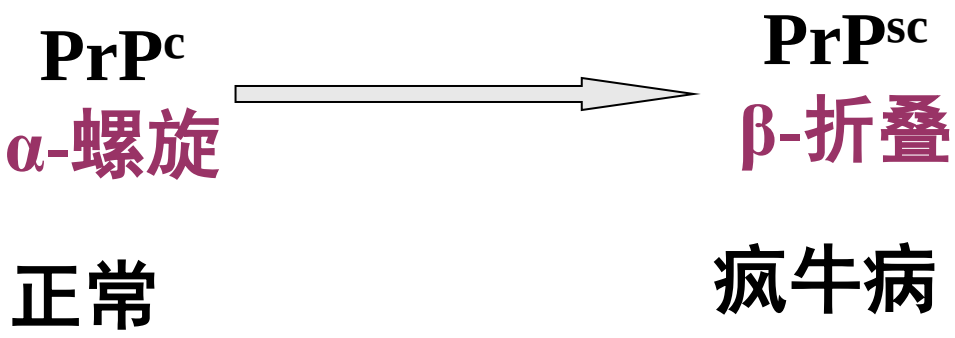
这类疾病包括：人纹状体脊髓变性病、老年痴呆症、亨停顿舞蹈病、疯牛病等。

■ 疯牛病中的蛋白质构象改变

疯牛病是由朊病毒蛋白(prion protein, PrP)引起的一组人和动物神经退行性病变。

正常的PrP富含α-螺旋，称为PrP^{c}。

PrP^{c}在某种未知蛋白质的作用下可转变成全为β-折叠的PrP^{sc}，从而致病。



第四节

蛋白质的理化性质

The Physical and Chemical Characters of Protein

一、蛋白质具有两性电离的性质

蛋白质分子除两端的氨基和羧基可解离外，氨基酸残基侧链中某些基团，在一定的溶液pH条件下都可解离成带负电荷或正电荷的基团。

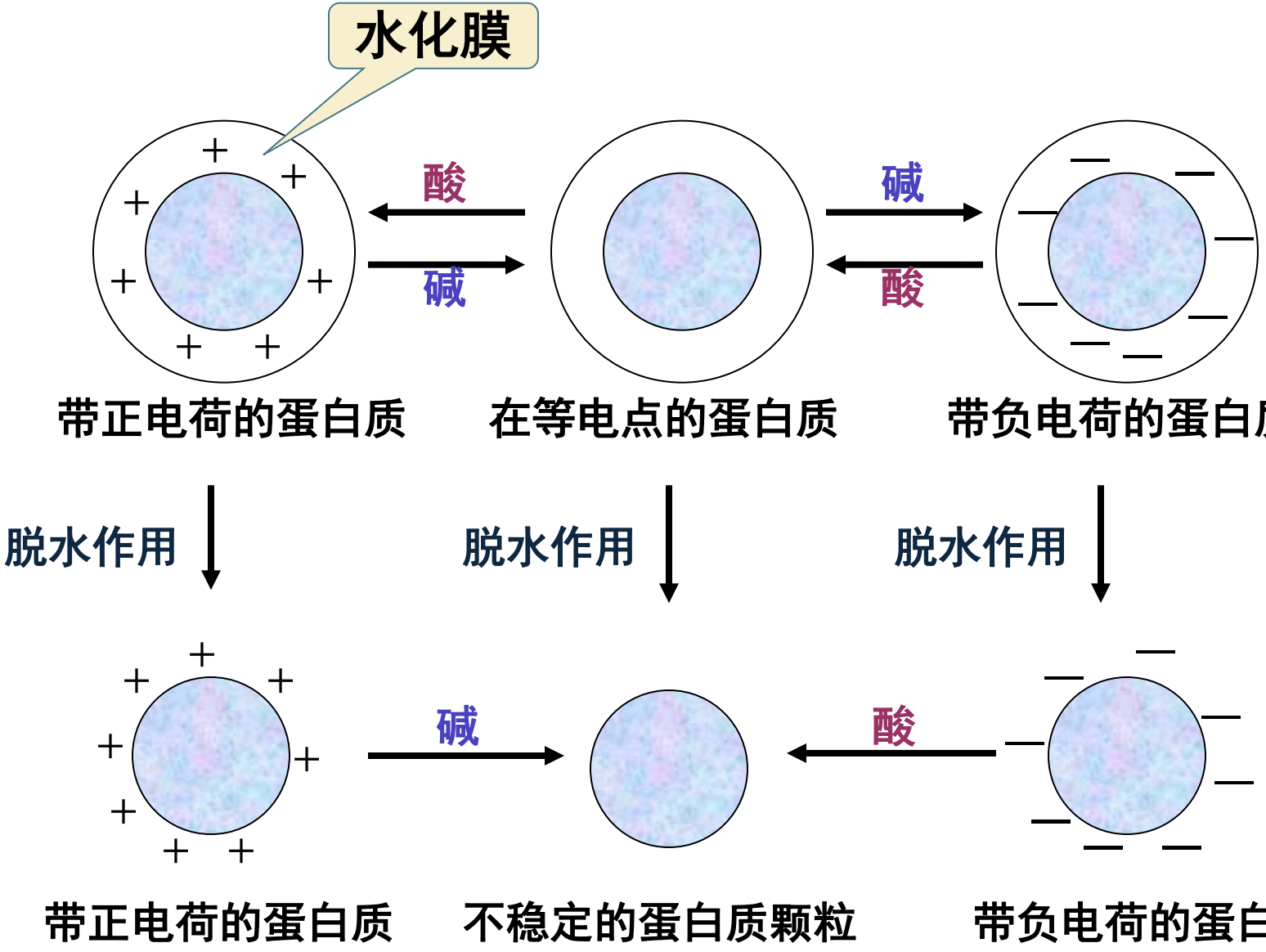
- **蛋白质的等电点(isoelectric point, pI)**

当蛋白质溶液处于某一pH时，蛋白质解离成正、负离子的趋势相等，即成为兼性离子，净电荷为零，此时溶液的pH称为蛋白质的等电点。

二、蛋白质具有胶体性质

蛋白质属于生物大分子之一，分子量可自1万至100万之巨，其分子的直径可达1～100nm，为胶粒范围之内。

- **蛋白质胶体的稳定因素:**
  - 表面同种电荷
  - 水化膜


水化膜
酸
碱
碱
酸
带正电荷的蛋白质
在等电点的蛋白质
带负电荷的蛋白质
脱水作用
脱水作用
脱水作用
碱
酸
带正电荷的蛋白质
不稳定的蛋白质颗粒
带负电荷的蛋白质

溶液中蛋白质的聚沉

三、蛋白质空间结构破坏而引起变性

- **蛋白质的变性(denaturation)**

在某些物理和化学因素作用下，其特定的空间构象被破坏，从而导致其理化性质改变和生物活性的丧失。

若蛋白质变性程度较轻，去除变性因素后，蛋白质仍可恢复或部分恢复其原有的构象和功能，称为复性(renaturation)。

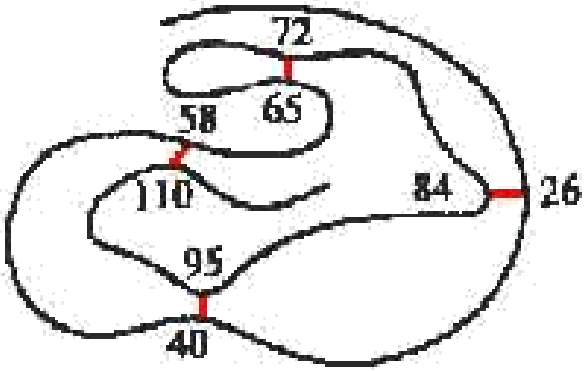
- **变性的本质：**

破坏非共价键和二硫键，不改变蛋白质的一级结构。

- **造成变性的因素：**

如加热、乙醇等有机溶剂、强酸、强碱、重金属离子及生物碱试剂等 。

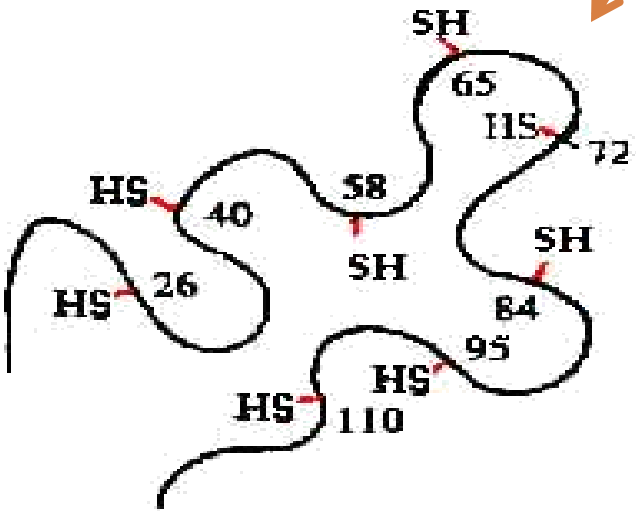
- **应用举例：**
  - **临床医学上，变性因素常被应用来消毒及灭菌。**
  - **此外，防止蛋白质变性也是有效保存蛋白质制剂（如疫苗等）的必要条件。**



天然状态，
有催化活性

去除尿素、
β-巯基乙醇

尿素、
β-巯基乙醇

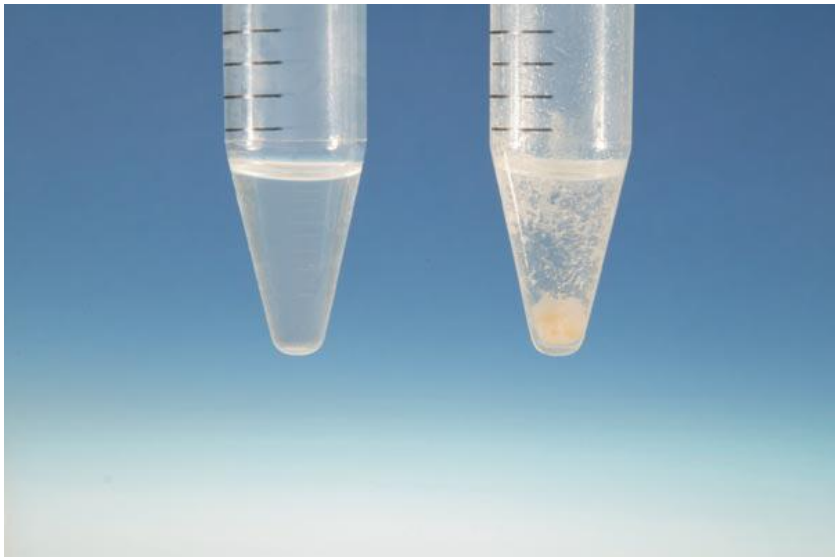


非折叠状态，无活性

What's precipitation of protein?

- 蛋白质分子相互聚集而从溶液中析出的现象称为沉淀(precipitation)

- 变性后的蛋白质由于疏水基团的暴露而易于沉淀，但沉淀的蛋白质不一定都是变性后的蛋白质。



What's coagulation of protein?

- 加热使蛋白质变性并凝聚成块状称为凝固（coagulation）
- 因此，凝固的蛋白质一定发生变性。

